POMPS

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About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

Contents

1 Signal Processing

1.1 Analog

1.1.1 A spring-mass-damper system with a mass of 1 kg is found to have a damping ratio of 0.2 and a natural frequency of 5 rad/s. The damping of the system is given by (AE 2007)

a) 2 Ns/m b) 2 N/s c) 0.2 kg/s d) 0.2 N/m

1.1.2 A 1 kg mass attached to a spring elongates it by 16 mm. The mass is then pulled from its equilibrium position by 10 mm and released from rest. Assuming the acceleration due to gravity of $9.81 m/s^2$, the response of the mass in mm is given by: (AE 2007)

a) $x = \cos 4.76\,t$ b) $x = \sin 4.76\,t$ c) $x = \sin 16\,t$ d) $x = 10\cos 16\,t$

1.1.3 A spring-mass-damper system is excited by a force $F_0 \sin \omega t$. The amplitude at resonance is measured to be 1 cm. At half the resonance frequency, the amplitude is 0.5 cm. The damping ratio of the system is (AE 2007)

a) 0.1026 b) 0.3242 c) 0.7211 d) 0.1936

1.1.4 $\lim_{\mathrm{x}\to 0} \frac{\sin \mathrm{x}}{\mathrm{e}^4 \mathrm{x}} =$ (AE 2007)

a) 10 b) 0 c) 1 d) ∞

1.1.5 Let a dynamical system be described by the differential equation $2\frac{dx}{dt} + \cos x = 0$. Which of the following differential equations describes this system in a close approximation sense for small perturbation about $x = \pi/4$? (AE 2007)

a) $2\frac{dx}{dt} + \sin x = 0$
b) $2\frac{dx}{dt} - \frac{1}{\sqrt{2}}x = 0$
c) $\frac{dx}{dt} + \cos x = 0$
d) $\frac{dx}{dt} + x = 0$

1.1.6 Consider the spring mass system shown in the figure below. This system has two degrees of freedom representing the motions of the two masses. (AE 2007)

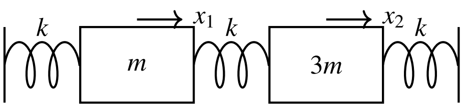


Fig. 1.1.6.1

a) The system shows the following type of coordinate coupling
   i) Static coupling
   ii) Dynamic coupling
   iii) Static and dynamic coupling
   iv) No coupling
b) The two natural frequencies of the system are given as

i) $\sqrt{\frac{4\pm\sqrt{5}k}{3m}}$ ii) $\sqrt{\frac{4\pm\sqrt{3}k}{3m}}$ iii) $\sqrt{\frac{4\pm\sqrt{7}k}{3m}}$ iv) $\sqrt{\frac{4\pm\sqrt{11}k}{3m}}$

1.1.7 Let

$$F(s) = \frac{(s+10)}{(s+2)(s+20)}$$

(AE 2007)

a) The partial fraction expansion of $F(s)$ is

i) $\frac{1}{s+2} + \frac{1}{s+20}$ ii) $\frac{5}{s+2} + \frac{2}{s+20}$ iii) $\frac{2}{s+2} + \frac{20}{s+20}$ iv) $\frac{4/9}{s+2} + \frac{5/9}{s+20}$

b) The inverse Laplace transform of $F(s)$ is

i) $2e^{-34} + 20e^{-34}$
ii) $\frac{4}{9}e^{-34} + \frac{5}{9}e^{-34}$
iii) $5e^{-34} + 2e^{-34}$
iv) $\frac{9}{4}e^{-34} + \frac{9}{5}e^{-34}$

1.1.8

$$\frac{\omega}{(s+a)^2 + \omega^2}$$

is the Laplace Transform of (AG 2007)

a) $\exp(-at)\sin\omega t$
b) $\sin\omega t$
c) $\exp(-at)\sinh\omega t$
d) $\sinh\omega t$

1.1.9 Evaluate $\int_0^{\pi} \frac{\sin t}{t} dt$ (CE 2007)

a) π b) $\pi/2$ c) $\pi/4$ d) $\pi/8$

1.1.10 The RC circuit shown is:

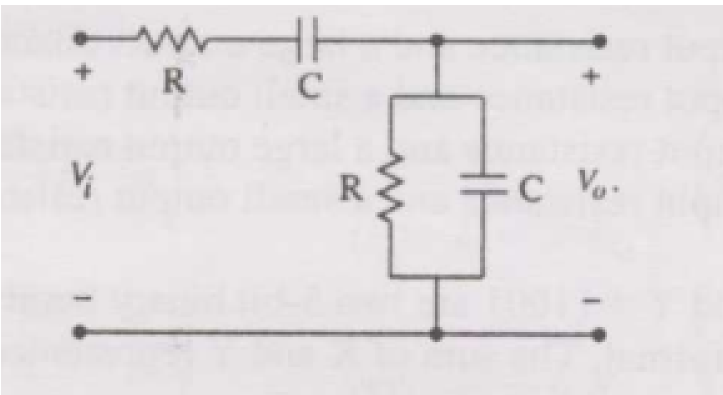


Fig. 1.1.10.1

(EC 2007)

a) Low-pass filter c) Band-pass filter ter
b) High-pass filter d) Band-reject fil-

1.1.11 The correct full wave rectifier circuit is:

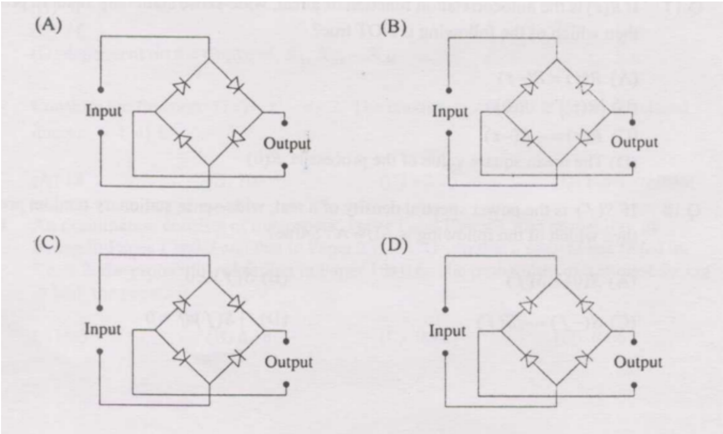


Fig. 1.1.11.1

1.1.12 Given closed-loop transfer function $T(s) = \frac{s-5}{(s+2)(s+3)}$, the system is: (EC 2007)
a) Unstable
b) Uncontrollable
c) Minimum phase
d) Non-minimum phase

1.1.13 If Laplace transform of $y(t)$ is $Y(s) = \frac{1}{s(s-1)}$, the final value is: (EC 2007)

a) -1 b) 0 c) 1 d) Unbounded

1.1.14 The solution of the differential equation $k^2 \frac{d^2y}{dx^2} = y$, under the boundary conditions (i) $y = y_1$ at $x = 0$ and (ii) $y = y_2$ at $x = \infty$, where k, y_1, y_2 are constants, is: (EC 2007)

(A) $y = (y_1 - y_2)\exp(-x/k) + y_2$
(B) $y = (y_1 - y_2)\exp(-x/k) + y_1$
(C) $y = (y_1 - y_2)\sinh(x/k) + y_2$
(D) $y = (y_1 - y_2)\exp(-x/k) + y_1$

1.1.15 Two series resonant filters are as shown in the figure. Let the 3-dB bandwidth of Filter 1 be B_1 and that of Filter 2 be B_2. The value of $\frac{B_1}{B_2}$ is:
(EC 2007)

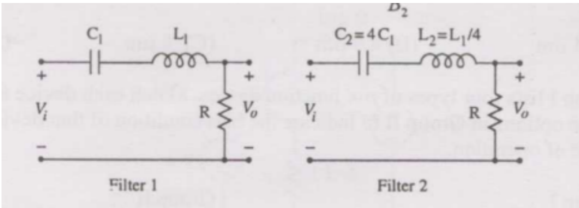


Fig. 1.1.15.1

(A) 4 (B) 1 (C) $\frac{1}{2}$ (D) $\frac{1}{4}$

1.1.16 In the circuit shown, V_C is 0 volts at $t = 0$ sec. For $t > 0$, the capacitor current $i_c(t)$, where t is in seconds, is given by: (EC 2007)

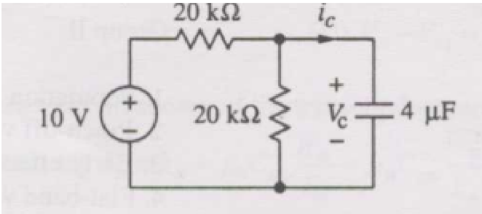


Fig. 1.1.16.1

a) $0.50\exp(-25t)$ mA
b) $0.25\exp(-25t)$ mA
c) $0.50\exp(-12.5t)$ mA
d) $0.25\exp(-6.25t)$ mA

1.1.17 The 3-dB bandwidth of the low-pass signal $e^{-t}u(t)$ is: (EC 2007)

a) $\frac{1}{2\pi}$ Hz b) $\frac{1}{2\pi}\sqrt{2}-1$ Hz c) ∞ d) 1 Hz

1.1.18 A Hilbert transformer is a: (EC 2007)

a) Non-linear system
b) Non-causal system
c) Time-varying system
d) Low-pass system

1.1.19 The frequency response of a linear time-invariant system is given by $H(f) = \frac{5}{1+j10\pi f}$. The step response of the system is: (EC 2007)

a) $5\left(1-e^{-5t}\right)u(t)$
b) $5\left(1-e^{-t/5}\right)u(t)$
c) $\frac{1}{5}\left(1-e^{-5t}\right)u(t)$
d) $\frac{1}{5}\left(1-e^{-t/5}\right)u(t)$

1.1.20 A control system with a PD controller is shown. If the velocity error constant $K_v = 1000$ and the damping ratio $\zeta = 0.5$, then the values of K_P and K_D are:

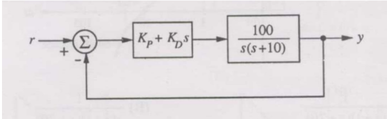


Fig. 1.1.20.1

(EC 2007)

a) $K_P = 100,\ K_D = 0.09$
b) $K_P = 100,\ K_D = 0.9$
c) $K_P = 10,\ K_D = 0.09$
d) $K_P = 10,\ K_D = 0.9$

1.1.21 The transfer function of a plant is $T(s) = \frac{5}{(s+5)(s^2+s+1)}$. The second-order approximation of $T(s)$ using the dominant pole concept is: (EC 2007)

(A) $\frac{1}{(s+5)(s+1)}$ (B) $\frac{5}{(s+5)(s+1)}$ (C) $\frac{5}{s^2+s+1}$ (D) $\frac{1}{s^2+s+1}$

1.1.22 The open-loop transfer function of a plant is given as $G(s) = \frac{1}{s^2-1}$. If the plant is operated in unity feedback, then the lead compensator that can stabilize the system is: (EC 2007)

a) $\frac{10(s-1)}{s+2}$ b) $\frac{10(s+4)}{s+2}$ c) $\frac{10(s+2)}{s+10}$ d) $\frac{2(s+2)}{s+10}$

1.1.23 A unity feedback control system has an open-loop transfer function $G(s) = \frac{K}{s(s^2+7s+12)}$. The gain K for which $s = -1 + j1$ will lie on the root locus of this system is: (EC 2007)

a) 4 b) 5.5 c) 6.5 d) 10

1.1.24 The asymptotic Bode magnitude plot of a transfer function is as shown: starts at 60 dB, slopes −20 dB/decade until 1 rad/s (at 40 dB), then −40 dB/decade crossing 0 dB near 10, then −60 dB/decade. The transfer function $G(s)$ corresponding to this plot is: (EC 2007)

a) $\frac{1}{(s+1)(s+20)}$ b) $\frac{1}{s(s+1)(s+20)}$ c) $\frac{100}{s(s+1)(s+20)}$ d) $\frac{100}{s(s+1)(1+0.05s)}$

1.1.25 The state-space representation of a separately excited DC servo motor dynamics is given as $\frac{d}{dt}\begin{pmatrix}\omega \\ i_a\end{pmatrix} = \begin{pmatrix}-1 & 1 \\ -1 & -10\end{pmatrix}\begin{pmatrix}\omega \\ i_a\end{pmatrix} + \begin{pmatrix}0 \\ 10\end{pmatrix}u$, where ω is the speed, i_a is the armature current, and u is the armature voltage.
The transfer function $\frac{\omega(s)}{u(s)}$ is: (EC 2007)

a) $\frac{10}{s^2+11s+11}$ b) $\frac{1}{s^2+11s+11}$ c) $\frac{10s+10}{s^2+11s+11}$ d) $\frac{1}{s^2+s+1}$

1.1.26 The raised cosine pulse $p(t)$ used for zero ISI with unity roll-off factor is

$$p(t) = \frac{\sin 4\pi Wt}{4\pi Wt(1 - 16W^2t^2)}.$$

The value of $p(t)$ at $t = \frac{1}{4W}$ is: (EC 2007)

a) −0.5 b) 0 c) 0.5 d) ∞

1.1.27 Consider the op-amp circuit shown.

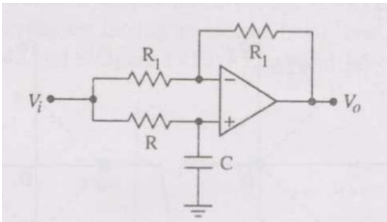


Fig. 1.1.27.1

a) The transfer function $V_o(s)/V_i(s)$ is: (EC 2007)

i) $\frac{1-sRC}{1+sRC}$ ii) $\frac{1+sRC}{1-sRC}$ iii) $\frac{1}{1-sRC}$ iv) $\frac{1}{1+sRC}$

b) If $V_i = V_i \sin(\omega t)$ and $V_o = V_o \sin(\omega t + \phi)$, the minimum and maximum values of ϕ (in radians) are respectively: (EC 2007)

i) $-\pi/2$ and $\pi/2$ ii) 0 and $\pi/2$ iii) $-\pi$ and 0 iv) $-\pi/2$ and 0

1.1.28 The system shown in the figure is (EE 2007)

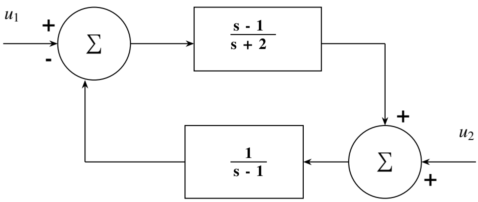


Fig. 1.1.28.1

a) stable
b) unstable
c) conditionally stable
d) stable for input u_1,but unstable for input u_2

1.1.29 Let a signal $a_1 \sin(\omega_1 t + \phi_1)$ be applied to a stable linear time-invariant system. Let the corresponding steady state output be represented as $a_2 F(\omega_1 t + \phi_2)$. Then which of the following statements is true? (EE 2007)

a) F is not necessarily a "sine" or "cosine" function but must be periodic with $\omega_1 = \omega_2$
b) F must be a "sine" or "cosine" function with $a_1 = a_2$
c) F must be a "sine" function with $\omega_1 = \omega_2$ and $\phi_1 = \phi_2$
d) F must be a "sine" or "cosine" function with $\omega_1 = \omega_2$

1.1.30 The frequency spectrum of a signal is shown in the figure. If this signal is ideally sampled at intervals of 1 ms, then the frequency spectrum of the sampled signal will be (EE 2007)

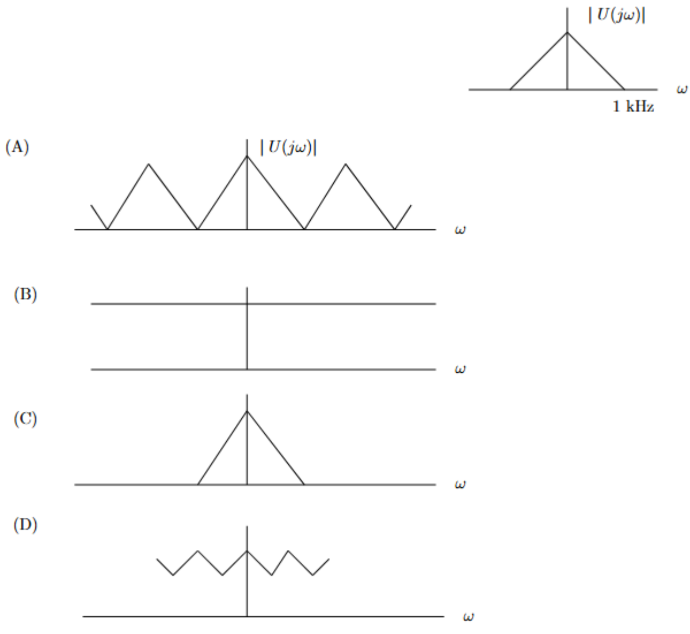


Fig. 1.1.30.1

1.1.31 The probes of a non-isolated, two-channel oscilloscope are clipped to points A, B and C in the circuit of the adjacent figure. V_{in} is a square wave of a suitable low frequency. The display on Ch_1 and Ch_2 are as shown on the right. Then the "Signal" and "Ground" probes S_1, G_1 and S_2, G_2 of Ch_1 and Ch_2 respectively are connected to points: (EE 2007)

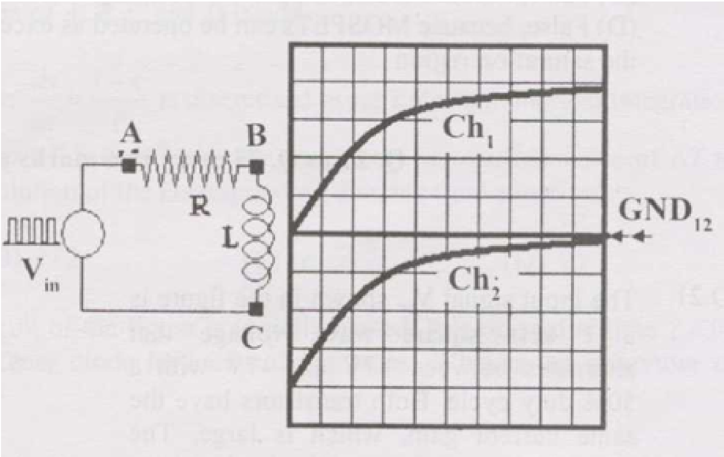


Fig. 1.1.31.1

a) A,B,C,A b) A,B,C,B c) C,B,A,B d) B,A,B,C

1.1.32 If $x = \text{Re } G(j\omega)$, and $y = \text{Im } G(j\omega)$ then for $\omega \to 0^+$, the Nyquist plot for

$$G(s) = \frac{1}{s(s+1)(s+2)}$$

becomes asymptotic to the line (EE 2007)

a) $x = 0$ b) $x = -3/4$ c) $x = y - 1/6$ d) $x = y/\sqrt{3}$

1.1.33 The system $\frac{900}{s(s+1)(s+9)}$ is to be compensated such that its gain-crossover frequency becomes same as its uncompensated phase-crossover frequency and provides a 45° phase margin. To achieve this, one may use (EE 2007)

a) a lag compensator that provides an attenuation of 20 dB and a phase lag of 45° at the frequency of $3\sqrt{3}$ rad/s
b) a lead compensator that provides an amplification of 20 dB and a phase lead of 45° at the frequency of 3 rad/s
c) a lag-lead compensator that provides an amplification of 20 dB and a phase lag of 45° at the frequency of $\sqrt{3}$ rad/s
d) a lag-lead compensator that provides an attenuation of 20 dB and phase lead of 45° at the frequency of 3 rad/s

1.1.34 A signal $x(t)$ is given by

$$x(t) = \begin{cases} 1, & -\frac{T}{4} < t \le \frac{3T}{4} \\ -1, & \frac{3T}{4} < t \le \frac{7T}{4} \\ -x(t+T) \end{cases}$$

Which among the following gives the fundamental Fourier term of $x(t)$? (EE 2007)

a) $\frac{4}{\pi}\cos\left(\frac{\pi t}{T} - \frac{\pi}{4}\right)$ c) $\frac{4}{\pi}\sin\left(\frac{\pi t}{T} - \frac{\pi}{4}\right)$
b) $\frac{\pi}{4}\cos\left(\frac{\pi t}{2T} + \frac{\pi}{4}\right)$ d) $\frac{\pi}{4}\sin\left(\frac{\pi t}{2T} + \frac{\pi}{4}\right)$

1.1.35 In the circuit shown in figure, switch SW_1 is initially CLOSED and SW_2 is OPEN. The inductor L carries a current of 10 A and the capacitor is charged to 10 V with polarities as indicated. SW_2 is initially CLOSED at $t = 0-$ and SW_1 is OPENED at $t = 0$. The current through C and the voltage across L at $t = 0+$ is (EE 2007)

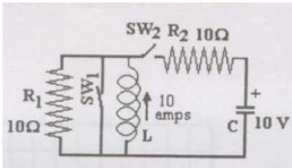


Fig. 1.1.35.1

a) 55A, 4.5V b) 5.5A, 45V c) 45A, 5.5V d) 4.5A, 55V

1.1.36 The R-L-C series circuit shown is supplied from a variable frequency voltage source. The admittance-locus of the R-L-C network at terminals AB for increasing frequency ω is (EE 2007)

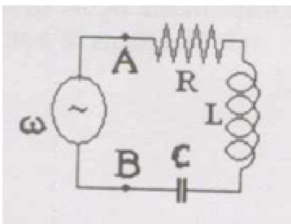
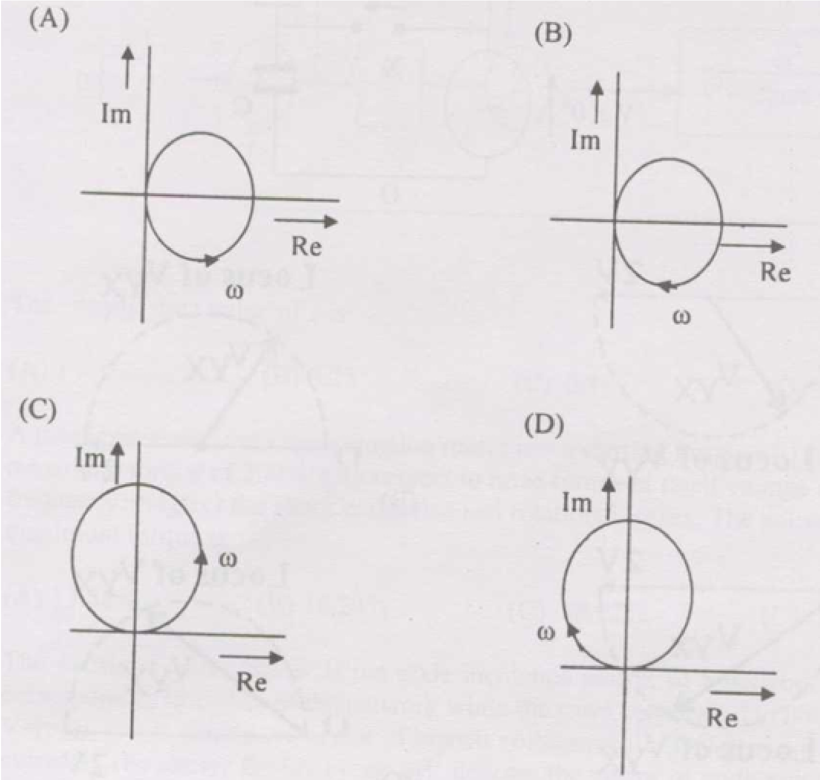


Fig. 1.1.36.1



1.1.37 If the loop gain K of a negative feedback system having a loop transfer function $\frac{K(s+3)}{(s+8)^2}$ is to be adjusted to induce a sustained oscillation then (EE 2007)

a) The frequency of this oscillation must be $\frac{4}{\sqrt{3}}$ rad/s
b) The frequency of this oscillation must be 4 rad/s
c) The frequency of this oscillation must be 4 or $\frac{4}{\sqrt{3}}$ rad/s
d) such a K does not exist

1.1.38 The system shown in figure below (EE 2007)

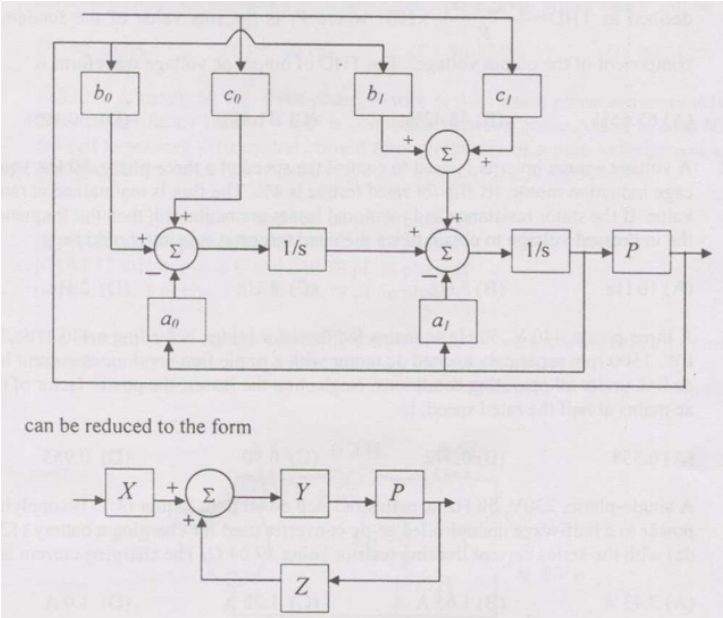


Fig. 1.1.38.1

with

a) $X = c_0 s + c_1, Y = \frac{1}{s^2+a_0 s+a_1}, Z = b_0 s + b_1$

b) $X = 1, Y = \frac{c_0 s+c_1}{s^2+a_0 s+a_1}, Z = b_0 s + b_1$

c) $X = c_1 s + c_0, Y = \frac{b_1 s+b_0}{s^2+a_1 s+a_0}, Z = 1$

d) $X = c_1 s + c_0, Y = \frac{1}{s^2+a_1 s+a_0}, Z = b_1 s + b_0$

1.1.39 A single-phase voltage source inverter is controlled in a single pulse-width modulated mode with a pulse width of 150° in each half cycle. Total harmonic distortion is defined as THD $= \frac{\sqrt{V_{rms}^2 - V_1^2}}{V_1} \times 100$, where V_1 is the rms value of the fundamental component of the output voltage. The THD of output ac voltage waveform is (EE 2007)

a) 65.65%   b) 48.42%   c) 31.83%   d) 30.49%

1.1.40 The state equation for the current I_1 shown in the network shown below in terms of the voltage V_x and the independent source V, is given by (EE 2007)

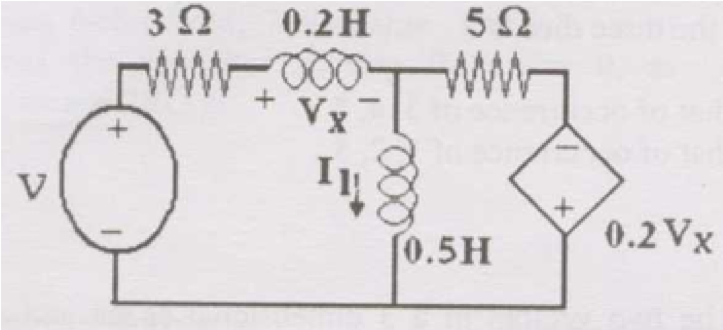


Fig. 1.1.40.1

a) $\frac{dI_1}{dt} = -1.4V_x - 3.75I_1 + \frac{5}{4}V$
b) $\frac{dI_1}{dt} = 1.4V_x - 3.75I_1 - \frac{5}{4}V$
c) $\frac{dI_1}{dt} = -1.4V_x + 3.75I_1 + \frac{5}{4}V$
d) $\frac{dI_1}{dt} = -1.4V_x + 3.75I_1 - \frac{5}{4}V$

1.1.41 If $u(t)$, $r(t)$ denote the unit step and unit ramp functions respectively and $u(t) * r(t)$ their convolution, then the function $u(t+1) * r(t-2)$ is given by (EE 2007)

a) $(1/2)(t-1)(t-2)$
b) $(1/2)(t-1)(t-2)$
c) $(1/2)(t-1)^2 u(t-1)$
d) none of the above

1.1.42 Consider the feedback control system shown below which is subjected to a unit step input. The system is stable and has the following parameters $k_p = 4$, $k_i = 10$, $\omega = 500$ and $\xi = 0.7$.

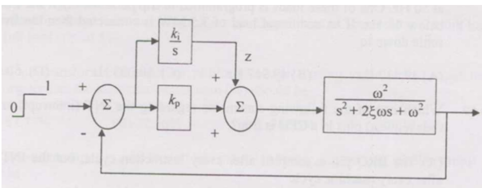


Fig. 1.1.42.1

The steady state value of the output is (EE 2007)

a) 1
b) 0.25
c) 0.1
d) 0

1.1.43 Consider the R-L-C circuit shown in figure.

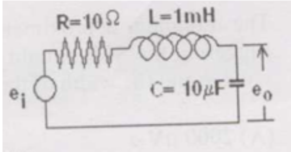


Fig. 1.1.43.1

a) For a step-input e_i, the overshoot in the output e_o will be (EE 2007)
   i) 0, since the system is not under-damped
   ii) 5%
   iii) 16%
   iv) 48%

b) If the above step response is to be observed on a non-storage CRO, then it would be best to have the e_i as a (EE 2007)
   i) step function
   ii) square wave of frequency 50 Hz
   iii) square wave of frequency 300 Hz
   iv) square wave of frequency 2.0 kHz

1.1.44 A signal is processed by a causal filter with transfer function $G(s)$. For a distortion free output signal waveform, $G(s)$ must (EE 2007)

a) provide zero phase shift for all frequency
b) provide constant phase shift for all frequency
c) provide linear phase shift that is proportional to frequency
d) provide a phase shift that is inversely proportional to frequency

1.1.45 $G(z) = \alpha z^{-1} + \beta z^{-3}$ is a low-pass digital filter with a phase characteristics same as that of the above question if (EE 2007)

a) $\alpha = \beta$ b) $\alpha = -\beta$ c) $\alpha = \beta^{(1/3)}$ d) $\alpha = \beta^{-(1/3)}$

1.1.46 Consider the AC bridge shown, with A, L and C having positive finite values.

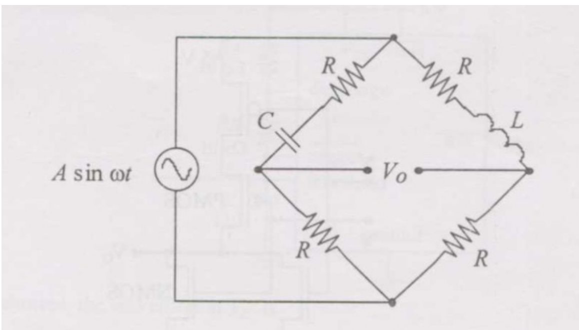


Fig. 1.1.46.1

Then (IN 2007)

a) $V_o = 0$ if $\omega L = \frac{1}{\omega C}$

b) $V_o = 0$ if $L = C$

c) $V_o = 0$ if $R = \frac{1}{\omega\sqrt{LC}}$

d) V_o cannot be made zero

1.1.47 A feedback control system with high gain K, is shown.

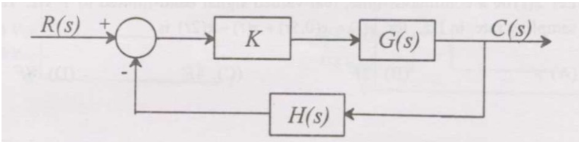


Fig. 1.1.47.1

Then the closed loop transfer function is (IN 2007)

a) sensitive to perturbations in $G(s)$ and $H(s)$

b) sensitive to perturbations in $G(s)$ but not to perturbations in $H(s)$

c) sensitive to perturbations in $H(s)$ but not to perturbations in $G(s)$

d) insensitive to perturbations in $G(s)$ and $H(s)$

1.1.48 Consider the following standard state-space description of a linear time-invariant single input single output system:

$$\dot{x} = Ax + Bu,$$
$$y = Cx + Du$$

Which one of the following statements about the transfer function **CANNOT** be true if $D \neq 0$? (IN 2007)

a) The system is unstable

b) The system is strictly proper

c) The system is low pass

d) The system is of type zero

1.1.49 The boundary-value problem

$$y'' + \lambda y = 0, \ y(0) = y(\pi) = 0$$

will have non-zero solutions if and only if the values of λ are (IN 2007)

a) $0, \pm1, \pm2, \ldots$

b) $1, 2, 3, \ldots$

c) $1, 4, 9, \ldots$

d) $1, 9, 25, \ldots$

1.1.50 In the circuit shown in the following figure, the current through the $1\ \Omega$ resistor is (IN 2007)

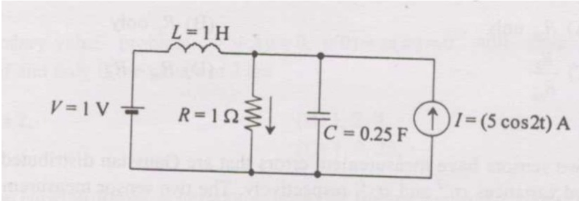


Fig. 1.1.50.1

a) $(1 + 5\cos 2t)$ A
b) $(5 + \cos 2t)$ A
c) $(1 - 5\cos 2t)$ A
d) 6 A

1.1.51 In the circuit shown in the following figure, the switch is kept closed for a long time and then opened at $t = 0$. (IN 2007)

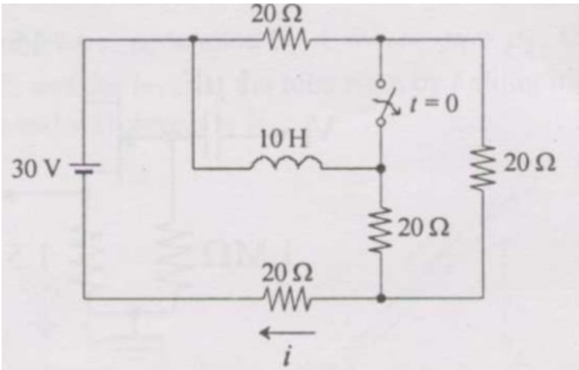


Fig. 1.1.51.1

The values of the current i just before opening the switch ($t = 0^-$) and just after opening the switch ($t = 0^+$) are, respectively

a) $\frac{3}{4}$ A and 1 A
b) $\frac{3}{4}$ A and $\frac{5}{4}$ A
c) 1 A and $\frac{7}{6}$ A
d) 1 A and 1 A

1.1.52 Consider the circuit shown below.

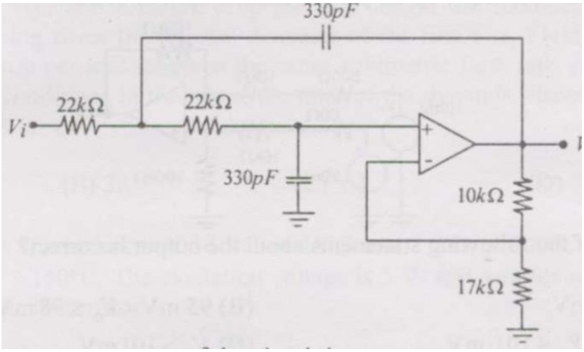


Fig. 1.1.52.1

The correct frequency response of the circuit is: (IN 2007)

a)

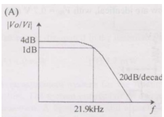


Fig. 1.1.52.2

b)

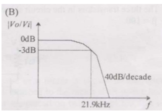


Fig. 1.1.52.3

c)

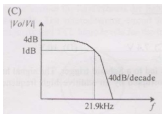


Fig. 1.1.52.4

d)

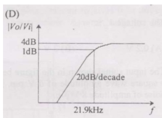


Fig. 1.1.52.5

1.1.53 In the circuit shown in the figure, the input signal is $v_i(t) = 5 + 3\cos(\omega t)$.

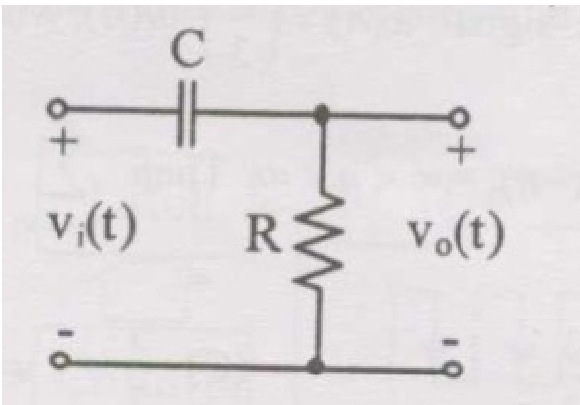


Fig. 1.1.53.1

The steady-state output is expressed as $v_o(t) = P + Q\cos(\omega t - \phi)$. If $\omega CR = 2$, the values of P and Q are: (IN 2007)

a) $P = 0,\ Q = \frac{6}{\sqrt{5}}$
b) $P = 0,\ Q = \frac{3}{\sqrt{5}}$
c) $P = 5,\ Q = \frac{6}{\sqrt{5}}$
d) $P = 5,\ Q = 3$

1.1.54 The signals $x(t)$ and $h(t)$ shown in the figures are convolved to yield $y(t)$.

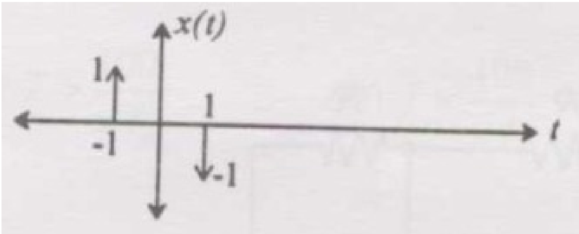


Fig. 1.1.54.1

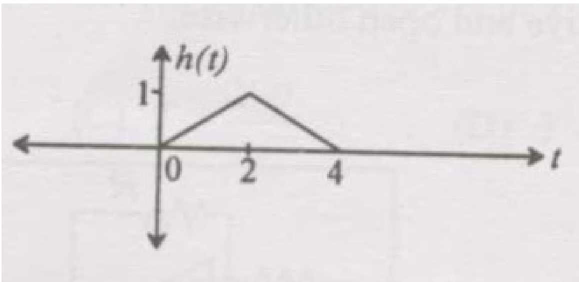


Fig. 1.1.54.2

Which one of the following figures represents the output $y(t)$? (IN 2007)

a)

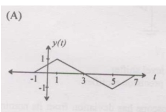


Fig. 1.1.54.3

b)

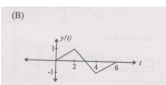


Fig. 1.1.54.4

c)

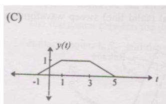


Fig. 1.1.54.5

d)

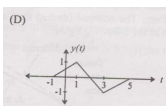


Fig. 1.1.54.6

1.1.55 Let the signal $x(t)$ have the Fourier transform $X(\omega)$. Consider the signal $y(t) = x(t-t_a)$ where t_a is an arbitrary delay. The magnitude of the Fourier transform of $y(t)$ is:

a) $|X(\omega)\cdot[\omega]|$ b) $|X(\omega)|\cdot\omega$ c) $\omega^2\cdot|X(\omega)|$ d) $|(\omega)||X(\omega)\cdot e^{-j\omega t_a}|$

1.1.56 Consider the triangular wave generator shown.

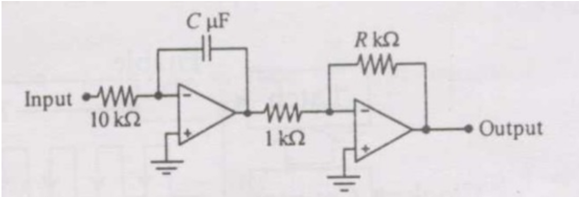


Fig. 1.1.56.1

Assume ideal op-amps and ±2 V supply. If the input is a +5 V, 50 Hz square wave with 50% duty cycle, the condition that results in a triangular wave of 5 V peak-to-peak and 50 Hz frequency at the output is:

(IN 2007)

a) $RC = 1$ b) $R/C = 1$ c) $R/C = 5$ d) $C/R = 5$

1.1.57 Two signals of peak-to-peak voltages 5 V and 2 V are fed to Channel 1 and Channel 2 of an oscilloscope. Vertical sensitivity is 1 V/div. The sine waves have identical frequency and phase. Trigger is manual at +1.25 V on Channel 1. Which trace correctly depicts Channel 2? (IN 2007)

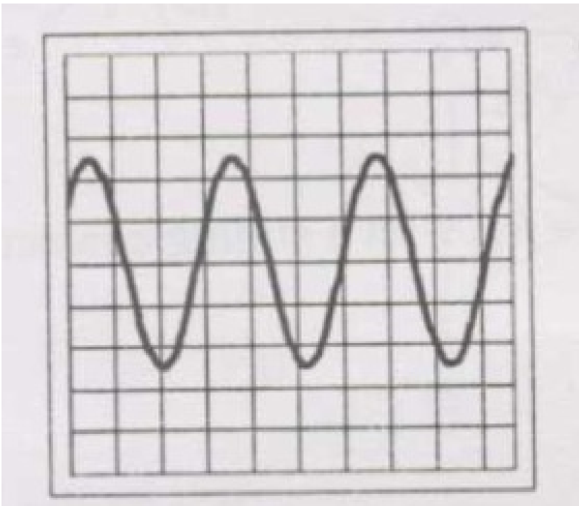


Fig. 1.1.57.1

a)

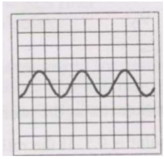


Fig. 1.1.57.2

c)

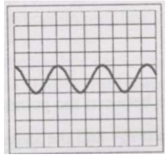


Fig. 1.1.57.4

b)

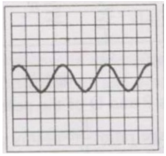


Fig. 1.1.57.3

d)

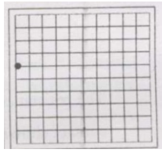


Fig. 1.1.57.5

1.1.58 A cascade control system with proportional controllers is shown.

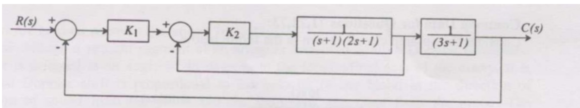


Fig. 1.1.58.1

Theoretically, the largest values of gains K_1 and K_2 that can be set without causing instability of the closed loop system are: (IN 2007)

a) 10 and 100 b) 100 and 10 c) 10 and 10 d) ∞ and ∞

1.1.59 The following figure represents a proportional control scheme of a first order system with transportation lag. (IN 2007)

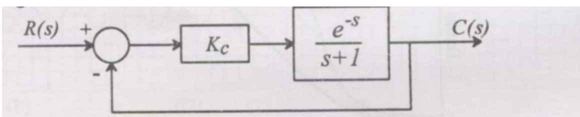


Fig. 1.1.59.1

a) The angular frequency in rad/s at which the loop phase lag becomes 180° is

i) 0.408 ii) 0.818 iii) 1.56 iv) 2.03

b) The steady state error for a unit step input when the gain $K_c = 1$ is

i) 1/4 ii) 1/2 iii) 1 iv) 2

1.1.60 A transfer function with unity DC gain has three poles at -1, -2, and -3 and no finite zeros. A plant with this transfer function is connected with a proportional controller of gain K in the forward path, in a unity feedback configuration. (IN 2007)

a) The transfer function is

i) $\frac{s}{(s-1)(s-2)(s-3)}$ iii) $\frac{s}{(s+1)(s+2)(s+3)}$
ii) $\frac{6}{(s+1)(s+2)(s+3)}$ iv) $\frac{6}{(s-1)(s-2)(s-3)}$

b) If the root locus plot of the closed loop system passes through the points $\pm j\sqrt{11}$, the maximum value of K for stability of the unity feedback closed loop system is

i) $\sqrt{11}$ ii) 6 iii) 10 iv) $\frac{6}{\sqrt{11}}$

1.1.61 If $F(s) = \tan^{-1}(s) + k$ is the Laplace transform of some function $f(t)$, $t \geq 0$, then $k =$ (MA 2007)

a) $-\pi$ b) $\frac{\pi}{2}$ c) 0 d) $\frac{\pi}{2}$

1.1.62 If $F(s)$ is the Laplace transform of function $f(t)$, then Laplace transform of $\int_0^t f(t)dt$ is (ME 2007)

a) $-F(s)$ b) $-F(s) - f(0)$ c) $\frac{F(s)}{s}$ d) $\int F(s)ds$

1.1.63 The natural frequency of the system shown below is (ME 2007)

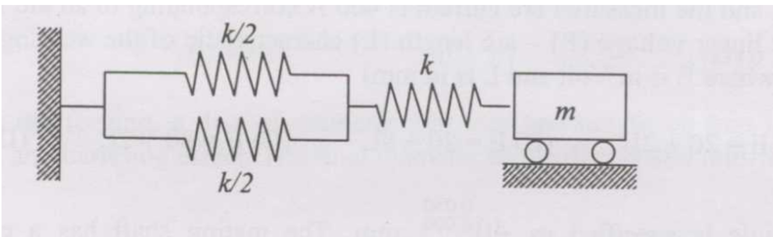


Fig. 1.1.63.1

a) $\sqrt{\frac{k}{2m}}$ b) $\sqrt{\frac{k}{m}}$ c) $\sqrt{\frac{2k}{m}}$ d) $\sqrt{\frac{3k}{m}}$

1.1.64 The equation of motion of a harmonic oscillator is given by (ME 2007)

$$\frac{dx^2}{dt^2} + 2\zeta\omega_n\frac{dx}{dt} + \omega_n^2 x = 0,$$

and the initial conditions at $t = 0$ are $x(0) = X$, $\frac{dx}{dt}(0) = 0$. The amplitude of $x(t)$ after n complete cycles is

a) $Xe^{-2\pi\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)}$
b) $Xe^{2\pi\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)}$
c) $Xe^{-2\pi\left(\frac{\sqrt{1-\zeta^2}}{\zeta}\right)}$
d) X

1.1.65 Inverse Laplace transform of $\frac{s+1}{s^2-4}$ is (PH 2007)

a) $\cos 2x + \frac{1}{2}\sin 2x$ b) $\cos x + \frac{1}{2}\sin x$ c) $\cosh x + \frac{1}{2}\sinh x$ d) $\cosh 2x + \frac{1}{2}\sinh 2x$

1.1.66 Assuming all components are ideal, the average power delivered by the DC voltage source in the network is:

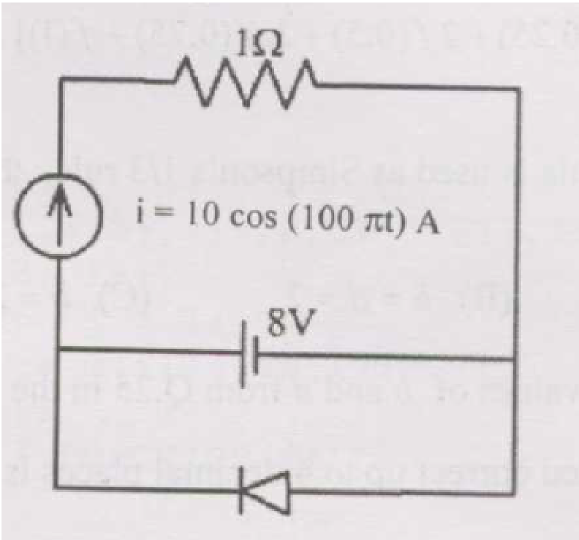


Fig. 1.1.66.1

(2007 XE)

a) -28W b) 0W c) 64W d) 80W

1.1.67 In the circuit below, $v(t) = 3\cos\omega t$, diode cut-in voltage 0.7 V, $V_1 = 2V$, $V_2 = 1V$. Max and min values of $v_o(t)$ are:

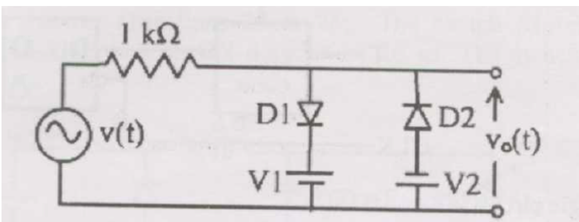


Fig. 1.1.67.1

(2007 XE)

a) +2.3 V, -1.7 V b) +2.7 V, -1.7 V c) +1.3 V, -0.3 V d) +2.3 V, -1.3 V

1.1.68 The switch S_1 turns on/off repeatedly at 20 kHz, with ON duration 20 μs. Switch & diode ideal. (2007 XE)

a) The average load voltage V_o is:

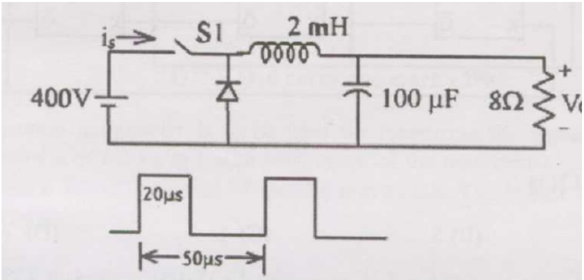


Fig. 1.1.68.1

i) 667 V ii) 400 V iii) 240 V iv) 160 V

b) Average source current I_s provided by 400 V source is:

i) 8 A ii) 12 A iii) 20 A iv) 160 A

1.1.69 The function defined by (AE 2008)

$$f(x) = \begin{cases} \sin x, & x < 0 \\ 0, & x = 0 \\ 3x^3, & x > 0 \end{cases}$$

a) is neither continuous nor differentiable at $x = 0$
b) is continuous and differentiable at $x = 0$
c) is differentiable but not continuous at $x = 0$
d) is continuous but not differentiable at $x = 0$

1.1.70 Which of the following is a solution of (AE 2008)

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = 0?$$

a) $e^{-x} + xe^{-x}$
b) $e^{x} + xe^{-x}$
c) $e^{t} + e^{x}$
d) $e^{-x} + xe^{x}$

1.1.71 Let $Y(s)$ denote the Laplace transform $\mathcal{L}(y(t))$ of the function $y(t) = \cosh(at)\sin(at)$. Then (AE 2008)

a) $\mathcal{L}\left(\frac{dy}{dt}\right) = \frac{dY}{ds}, \quad \mathcal{L}(t\,y(t)) = sY(s)$
b) $\mathcal{L}\left(\frac{dy}{dt}\right) = sY(s), \quad \mathcal{L}(t\,y(t)) = -\frac{dY}{ds}$
c) $\mathcal{L}\left(\frac{dy}{dt}\right) = \frac{dY}{ds}, \quad \mathcal{L}(t\,y(t)) = Y(s-1)$
d) $\mathcal{L}\left(\frac{dy}{dt}\right) = sY(s), \quad \mathcal{L}(t\,y(t)) = e^{a}Y(s)$

1.1.72 Inverse Laplace Transform of $\frac{1}{(s-2)^2}$ is (AG 2008)

a) e^{2t} b) te^{2t} c) $2te^{t}$ d) t^2e^{2t}

1.1.73 Solution of the differential equation (AG 2008)

$$\frac{dy}{dx} - 7y = e^{x}$$

is

a) $e^{x}(Ce^{6x} - 6^{-1})$ b) $e^{7x}(e^{-6x} - C)^{-1}$ c) $e^{x}(Ce^{-6x} - 6^{-1})$ d) $e^{x}[Ce^{x} + (6e^{x})^{-1}]$

1.1.74 The differential equation of motion for a single degree of freedom mass-spring damped system is

$$\frac{d^2x}{dt^2} + \frac{dx}{dt} + 12x = 0$$

If the units of mass, length and time are kg, m, and s respectively, the natural frequency of vibration is (AG 2008)

a) 0.42 rad s^{-1} b) 0.52 rad s^{-1} c) 1.22 rad s^{-1} d) 1.83 rad s^{-1}

1.1.75 Which ONE of the following transformations $\{u = f(y)\}$ reduces $\frac{dy}{dx} + Ay^3 + By = 0$ to a linear differential equation? (A and B are positive constants) (CH 2008)

a) $u = y^{-3}$ b) $u = y^{-2}$ c) $u = y^{-1}$ d) $u = y^{2}$

1.1.76 The Laplace transform of the function $f(t) = t\sin t$ is (CH 2008)

a) $\frac{2s}{(s^2+1)^2}$
b) $\frac{1}{s^2(s^2+1)}$
c) $\frac{1}{s^2} + \frac{1}{(s^2+1)}$
d) $\frac{1}{(s-1)^2+1}$

1.1.77 The unit impulse response of a first order process is given by $2e^{-0.5t}$. The gain and time constant of the process are, respectively, (CH 2008)

a) 4 and 2 b) 2 and 2 c) 2 and 0.5 d) 1 and 0.5

1.1.78 A unit step input is given to a process that is represented by the transfer function $\frac{(s+2)}{(s+5)}$. The initial value ($t = 0^{+}$) of the response of the process to the step input is (CH 2008)

a) 0 b) 2/5 c) 1 d) ∞

1.1.79 Which ONE of the following transfer functions corresponds to an inverse response process with a positive gain? (CH 2008)

a) $\frac{1}{2s+1} - \frac{2}{3s+1}$ b) $\frac{2}{s+1} - \frac{5}{s+10}$ c) $\frac{3(0.5s-1)}{(2s+1)(s+1)}$ d) $\frac{5}{s+1} - \frac{3}{2s+1}$

1.1.80 The cross-over frequency associated with a feedback loop employing a proportional controller to control the process represented by the transfer function

$$G_p(s) = \frac{2e^{-s}}{(\tau s + 1)^2}, \quad \text{(units of time is minutes)}$$

is found to be 0.6 rad/min. Assume that the measurement and valve transfer functions are unity. (CH 2008)

a) The time constant, τ (in minutes) is

i) 1.14 ii) 1.92 iii) 3.23 iv) 5.39

b) If the control loop is to operate at a gain margin of 2.0, the gain of the proportional controller must equal

i) 0.85 ii) 2.87 iii) 3.39 iv) 11.50

1.1.81 In the following circuit, the switch S is closed at $t = 0$. The rate of change of current $\frac{di}{dt}(0^+)$ is given by (EC 2008)

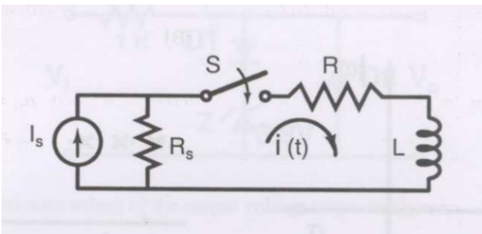


Fig. 1.1.81.1

a) 0 b) $\frac{R_s I_s}{L}$ c) $\frac{(R+R_s)I_s}{L}$ d) ∞

1.1.82 The input and output of a continuous time system are respectively denoted by $x(t)$ and $y(t)$. Which of the following descriptions corresponds to a causal system? (EC 2008)

a) $y(t) = x(t-2) + x(t+4)$
b) $y(t) = (t-4)x(t+1)$
c) $y(t) = (t+4)x(t-1)$
d) $y(t) = (t+5)x(t+5)$

1.1.83 The impulse response $h(t)$ of a linear time-invariant continuous time system is described by $h(t) = \exp(\alpha t)u(t) + \exp(\beta t)u(t)$, where $u(t)$ denotes the unit step function, and α and β are real constants. This system is stable if (EC 2008)

a) α is positive and β is positive
b) α is negative and β is negative
c) α is positive and β is negative
d) α is negative and β is positive

1.1.84 The pole-zero plot given below corresponds to a (EC 2008)

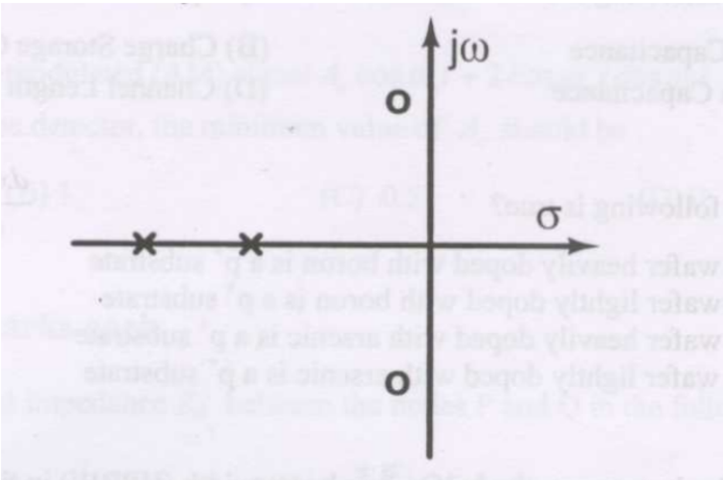


Fig. 1.1.84.1

a) Low pass filter b) High pass filter c) Band pass filter d) Notch filter

1.1.85 Step responses of a set of three second-order underdamped systems all have the same percentage overshoot. Which of the following diagrams represents the poles of the three systems? (EC 2008)

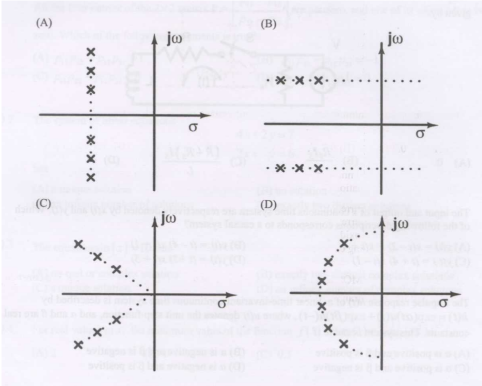


Fig. 1.1.85.1

a) (A) b) (B) c) (C) d) (D)

1.1.86 The circuit shown in the figure is used to charge the capacitor C alternately from two current sources as indicated. The switches S_1 and S_2 are mechanically coupled and connected as follows

$$S_1 \to P_1, S_2 \to P_2, \quad 2nT \leq t < (2n+1)T, \qquad n = 0, 1, 2, \ldots,$$
$$S_1 \to Q_1, S_2 \to Q_2, \quad (2n+1)T \leq t < (2n+2)T, \qquad n = 0, 1, 2, \ldots$$

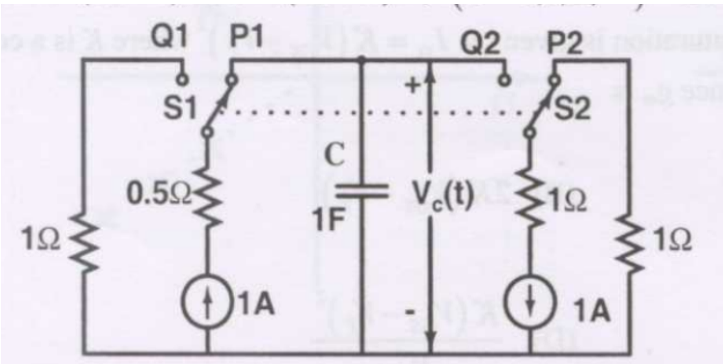


Fig. 1.1.86.1

Assume that the capacitor has zero initial charge. Given that $u(t)$ is a unit step function, the voltage $V_c(t)$ across the capacitor is given by (EC 2008)

a) $\sum_{n=0}^{\infty}(-1)^n t u(t - nT)$ b) $u(t) + 2\sum_{n=1}^{\infty}(-1)^n u(t - nT)$

c) $tu(t) + 2\sum_{n=1}^{\infty}(-1)^n(t - nT)u(t - nT)$ d) $\sum_{n=0}^{\infty}\left[0.5 - e^{-(t-2nT)} + 0.5e^{-(t-2nT-T)}\right]$

1.1.87 A linear, time-invariant, causal continuous time system has a rational transfer function with simple poles at $s = -2$ and $s = -4$, and one simple zero at $s = -1$. A unit step $u(t)$ is applied at the input of the system. At steady state, the output has constant value of 1. The impulse response of this system is (EC 2008)

a) $[\exp(-2t) + \exp(-4t)]\, u(t)$
b) $[-4\exp(-2t) + 12\exp(-4t) - \exp(-t)]\, u(t)$
c) $[-4\exp(-2t) + 12\exp(-4t)]\, u(t)$
d) $[-0.5\exp(-2t) + 1.5\exp(-4t)]\, u(t)$

1.1.88 The signal $x(t)$ is described by

$$x(t) = \begin{cases} 1 & \text{for } -1 \le t \le +1 \\ 0 & \text{otherwise} \end{cases}$$

Two of the angular frequencies at which its Fourier transform becomes zero are (EC 2008)

a) π, 2π b) 0.5π, 1.5π c) 0, π d) 2π, 2.5π

1.1.89 Let $x(t)$ be the input and $y(t)$ be the output of a continuous time system. Match the system properties P1, P2, and P3 with system relations R1, R2, R3, R4. (EC 2008)

Properties	Relations
P1: Linear but NOT time-invariant	R1: $y(t) = t^2 x(t)$
P2: Time-invariant but NOT linear	R2: $y(t) = \lvert x(t)\rvert$
P3: Linear and time-invariant	R3: $y(t) = x(t)$
	R4: $y(t) = x(t - 5)$

Fig. 1.1.89.1

a) (P1, R1), (P2, R3), (P3, R4)
b) (P1, R2), (P2, R3), (P3, R4)
c) (P1, R3), (P2, R1), (P3, R2)
d) (P1, R1), (P2, R2), (P3, R3)

1.1.90 Group I lists a set of four transfer functions. Group II gives a list of possible step responses $y(t)$. Match the step responses with the corresponding transfer functions.

Group I

$P = \frac{25}{s^2+25}$ $Q = \frac{36}{s^2+20s+36}$ $R = \frac{36}{s^2+12s+36}$ $S = \frac{49}{s^2+7s+49}$

Group II

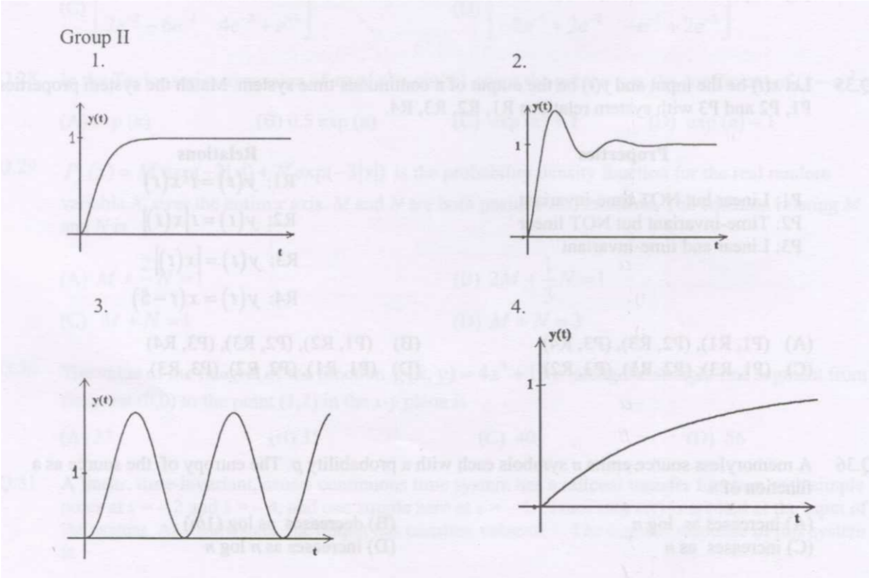


Fig. 1.1.90.1

(EC 2008)

a) P-3, Q-1, R-4, S-2
b) P-3, Q-2, R-4, S-1
c) P-2, Q-1, R-4, S-3
d) P-3, Q-4, R-1, S-2

1.1.91 A certain system has transfer function $G(s) = \frac{s+8}{s^2+as-4}$, where a is a parameter. Consider the standard negative unity feedback configuration.

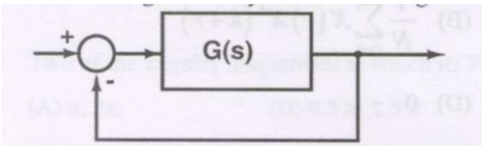


Fig. 1.1.91.1

Which of the following statements is true? (EC 2008)

a) The closed loop system is never stable for any value of a.
b) For some positive values of a, the closed loop system is stable, but not for all positive values.
c) For all positive values of a, the closed loop system is stable.
d) The closed loop system is stable for all values of a, both positive and negative.

1.1.92 A signal flow graph of a system is given below.

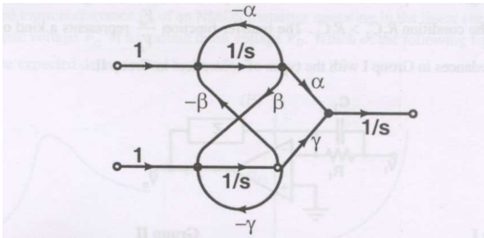


Fig. 1.1.92.1

The set of equations that correspond to this signal flow graph is (EC 2008)

a) $\frac{d}{dt}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = \begin{pmatrix}\beta & -\gamma & 0\\ \gamma & \alpha & 0\\ -\alpha & -\beta & 0\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} + \begin{pmatrix}1 & 0\\0 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

b) $\frac{d}{dt}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = \begin{pmatrix}0 & \alpha & -\gamma\\ -\gamma & 0 & \beta\\ \alpha & \beta & 0\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} + \begin{pmatrix}1 & 0\\0 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

c) $\frac{d}{dt}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = \begin{pmatrix}-\gamma & 0 & \beta\\ \alpha & \gamma & 0\\ -\beta & 0 & -\alpha\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} + \begin{pmatrix}1 & 0\\0 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

d) $\frac{d}{dt}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} = \begin{pmatrix}-\gamma & 0 & \beta\\ 0 & \alpha & 0\\ -\beta & 0 & -\alpha\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix} + \begin{pmatrix}1 & 0\\0 & 1\\0 & 0\end{pmatrix}\begin{pmatrix}u_1\\u_2\end{pmatrix}$

1.1.93 The number of open right half plane poles of $G(s) = \frac{10}{s^3+2s^4+3s^3+6s^2+5s+3}$ is (EC 2008)

a) 0 b) 1 c) 2 d) 3

1.1.94 The magnitude of frequency response of an underdamped second order system is 5 at 0 rad/sec and peaks to $\frac{10}{\sqrt{3}}$ at $5\sqrt{2}$ rad/sec. The transfer function of the system is (EC 2008)

a) $\frac{500}{s^2+10s+100}$ b) $\frac{375}{s^2+5s+75}$ c) $\frac{720}{s^2+12s+144}$ d) $\frac{1125}{s^2+25s+225}$

1.1.95 Group I gives two possible choices for the impedance Z in the diagram. The circuit elements in Z satisfy the condition $R_2C_2 > R_1C_1$. The transfer function $\frac{V_o}{V_i}$ represents a kind of controller. Match the impedances in Group I with the types of controllers in Group II. (EC 2008)

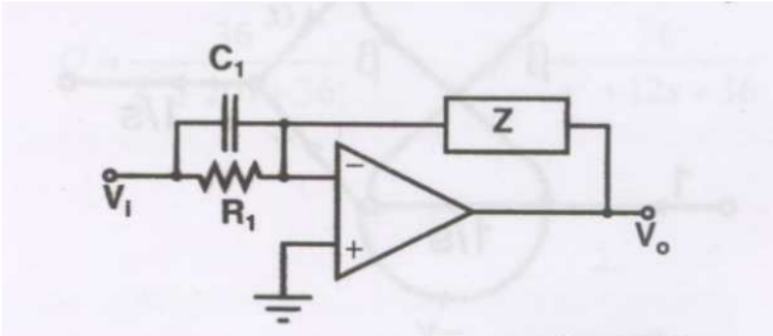


Fig. 1.1.95.1

Group I:

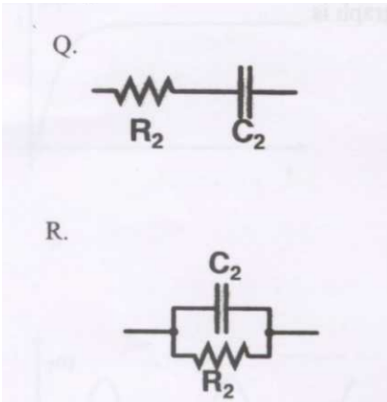


Fig. 1.1.95.2: Group I

Group II:
1. PID controller 2. Lead compensator 3. Lag compensator

a) Q-1, R-2 b) Q-1, R-3 c) Q-2, R-3 d) Q-3, R-2

1.1.96 The OPAMP circuit shown below represents a (EC 2008)

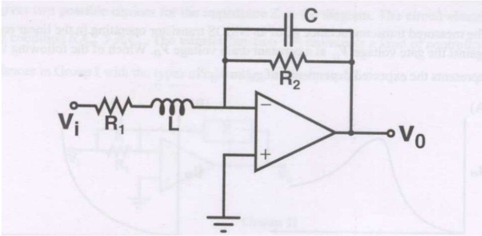


Fig. 1.1.96.1

a) high pass filter
b) low pass filter
c) band pass filter
d) band reject filter

1.1.97 Consider the frequency modulated signal $10\cos[2\pi \times 10^5 t + 5\sin(2\pi \times 1500t) + 7.5\sin(2\pi \times 1000t)]$ with carrier frequency of 10^5 Hz. The modulation index is (EC 2008)

a) 12.5
b) 10
c) 7.5
d) 5

1.1.98 The signal $\cos\Omega_1 t + 0.5\cos\Omega_f t\sin\Omega_t t$ is (EC 2008)

a) FM only
b) AM only
c) both AM and FM
d) neither AM nor FM

1.1.99 The following series RLC circuit with zero initial conditions is excited by a unit impulse function $\delta(t)$. (EC 2008)

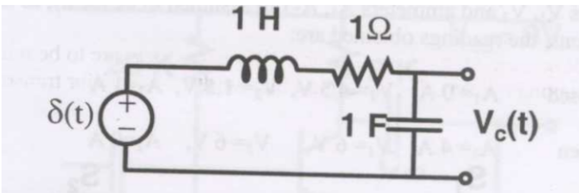


Fig. 1.1.99.1

a) For $t > 0$, the output voltage $V_c(t)$ is

i) $\frac{2}{\sqrt{3}}\left(e^{-\frac{1}{2}t} - e^{-\frac{\sqrt{3}}{2}t}\right)$
ii) $\frac{2}{\sqrt{3}}te^{-\frac{1}{2}t}$
iii) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right)$
iv) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)$

b) For $t > 0$, the voltage across the resistor is

i) $\frac{1}{\sqrt{3}}\left(e^{-\frac{\sqrt{3}}{2}t} - e^{-\frac{1}{2}t}\right)$
iii) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\sin\left(\frac{\sqrt{3}}{2}t\right)$
ii) $e^{-\frac{1}{2}t}\left[\cos\left(\frac{\sqrt{3}}{2}t\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}}{2}t\right)\right]$
iv) $\frac{2}{\sqrt{3}}e^{-\frac{1}{2}t}\cos\left(\frac{\sqrt{3}}{2}t\right)$

1.1.100 The impulse response $h(t)$ of a linear time-invariant continuous time system is given by $h(t) = \exp(-2t)u(t)$, where $u(t)$ denotes the unit step function. (EC 2008)

a) The frequency response $H(\Omega)$ of this system in terms of angular frequency Ω, is given by $H(\Omega) =$

i) $\frac{1}{1+j2\Omega}$
ii) $\frac{\sin\Omega}{\Omega}$
iii) $\frac{1}{2+j\Omega}$
iv) $\frac{j\Omega}{2+j\Omega}$

b) The output of this system to the sinusoidal input $x(t) = 2\cos(2t)$ for all time t is

i) 0
iii) $2^{-0.5}\cos(2t - 0.125\pi)$
ii) $2^{-0.25}\cos(2t - 0.125\pi)$
iv) $2^{-0.5}\cos(2t - 0.25\pi)$

1.1.101 A signal $e^{-\alpha t}\sin(\omega t)$ is the input to a real Linear Time Invariant system. Given K and ϕ are constants, the output of the system will be of the form $Ke^{-\beta t}\sin(\nu t + \phi)$ where (EE 2008)

a) β need not be equal to α but ν equal to ω
b) ν need not be equal to ω but β equal to α
c) β equal to α and ν equal to ω
d) β need not be equal to α and ν need not be equal to ω

1.1.102 The equivalent circuits of a diode, during forward biased and reverse biased conditions, are shown in the figure

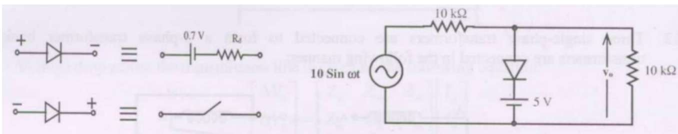


Fig. 1.1.102.1

If such a diode is used in clipper circuit of figure given above, the output voltage (v_0) of the circuit will be (EE 2008)

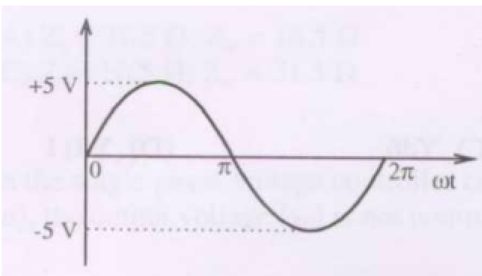


Fig. 1.1.102.2

a)

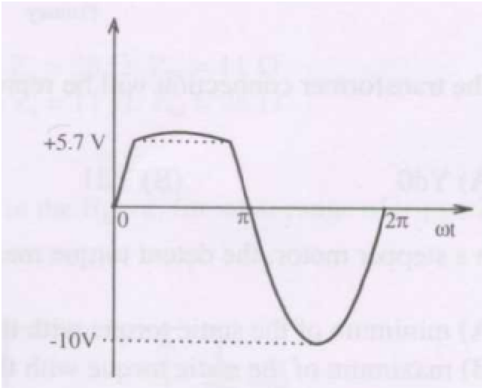


Fig. 1.1.102.3

b)

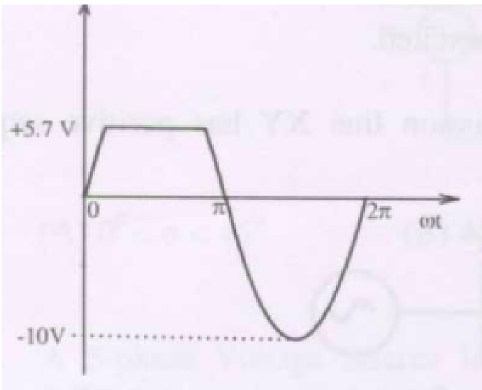


Fig. 1.1.102.4

c)

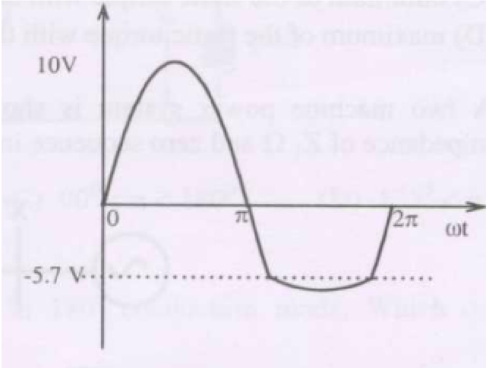


Fig. 1.1.102.5

d)

1.1.103 The impulse response of a causal linear time-invariant system is given as $h(t)$. Now consider the following two statements:
Statement (I): Principle of superposition holds
Statement (II): $h(t) = 0$ for $t < 0$.

Which one of the following statements is correct? (EE 2008)

a) Statement (I) is correct and Statement (II) is wrong
b) Statement (II) is correct and Statement (I) is wrong
c) Both Statement (I) and Statement (II) are wrong
d) Both Statement (I) and Statement (II) are correct

1.1.104 The time constant for the given circuit will be (EE 2008)

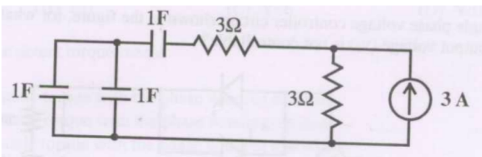


Fig. 1.1.104.1

a) 1/9 s b) 1/4 s c) 4 s d) 9 s

1.1.105 The resonant frequency for the given circuit will be (EE 2008)

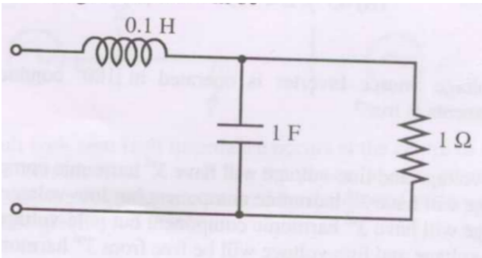


Fig. 1.1.105.1

a) 1 rad/s b) 2 rad/s c) 3 rad/s d) 4 rad/s

1.1.106 A system with input $x(t)$ and output $y(t)$ is defined by the input-output relation:

$$y(t) = \int_{-\infty}^{-2t} x(\tau)\, d\tau$$

The system will be (EE 2008)

a) causal, time-invariant and unstable
b) causal, time-invariant and stable
c) non-causal, time-invariant and unstable
d) non-causal, time-variant and unstable

1.1.107 A signal $x(t) = \text{sinc}(\alpha t)$ where α is a real constant $\left(\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}\right)$ is the input to a Linear Time invariant system whose impulse response $h(t) = \text{sinc}(\beta t)$ where β is a real constant. If $\min(\alpha,\beta)$ denotes the minimum of α and β, and similarly $\max(\alpha,\beta)$ denotes the maximum of α and β, and K is a constant, which one of the following statements is true about the output of the system? (EE 2008)

a) It will be of the form $K\ \text{sinc}(\gamma t)$ where $\gamma = \min(\alpha,\beta)$
b) It will be of the form $K\ \text{sinc}(\gamma t)$ where $\gamma = \max(\alpha,\beta)$
c) It will be of the form $K\ \text{sinc}(\alpha t)$
d) It cannot be a sinc type of signal

1.1.108 Let $x(t) = \text{rect}\left(t - \frac{1}{2}\right)$ (where $\text{rect}(x) = 1$ for $-\frac{1}{2} \le x \le \frac{1}{2}$ and zero otherwise). Then if $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$, the Fourier Transform of $x(t) + x(-t)$ will be given by (EE 2008)

a) $\text{sinc}\left(\frac{\omega}{2\pi}\right)$ b) $2\ \text{sinc}\left(\frac{\omega}{2\pi}\right)$ c) $2\ \text{sinc}\left(\frac{\omega}{2\pi}\right)\cos\left(\frac{\omega}{2}\right)$ d) $\text{sinc}\left(\frac{\omega}{2\pi}\right)\sin\left(\frac{\omega}{2}\right)$

(EE 2008)

1.1.109 In the voltage doubler circuit shown in the figure, the switch 'S' is closed at t = 0. Assuming diodes D_1 and D_2 to be ideal, load resistance to be infinite and initial capacitor voltages to be zero, the steady state voltage across capacitors C_1 and C_2 will be (EE 2008)

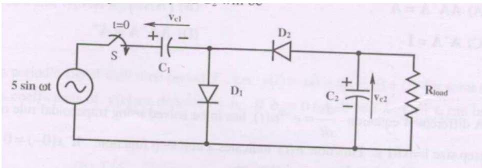


Fig. 1.1.109.1

a) $v_{C1} = 10\text{V}, v_{C2} = 5\text{V}$
b) $v_{C1} = 10\text{V}, v_{C2} = -5\text{V}$
c) $v_{C1} = 5\text{V}, v_{C2} = 10\text{V}$
d) $v_{C1} = 5\text{V}, v_{C2} = -10\text{V}$

1.1.110 A at waveform point 'P' with generator circuit using OPAMPs is shown in the figure. It produces a triangular wave a peak to peak voltage of 5 V for v; = 0 V

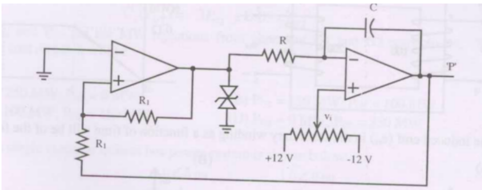


Fig. 1.1.110.1

If the voltage v; is made +2.5V, the voltage waveform at point 'P' will become (EE 2008)

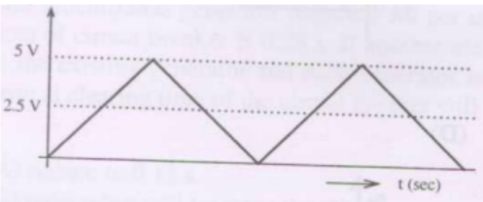


Fig. 1.1.110.2

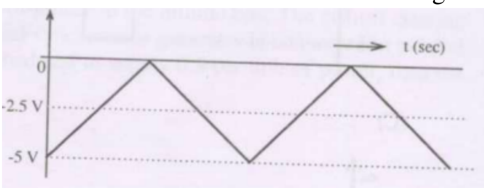


Fig. 1.1.110.3

a)

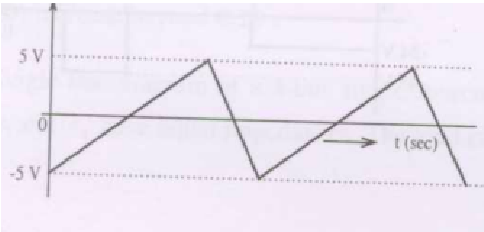


Fig. 1.1.110.4

b)

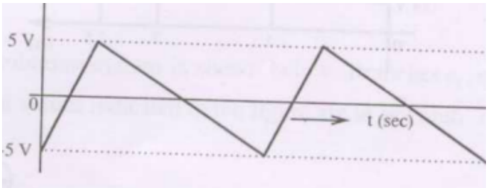


Fig. 1.1.110.5

c)

(EE 2008)

1.1.111 A single-phase half controlled converter shown in the figure is feeding power to highly inductive load. The converter is operating at a firing angle of 60°

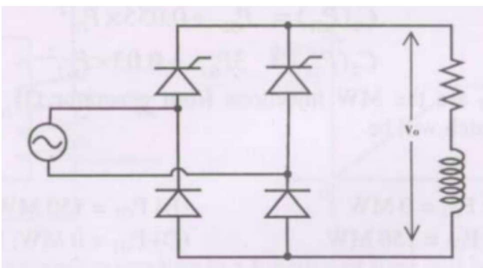


Fig. 1.1.111.1

If the firing pulses are suddenly removed, the steady state voltage v_o waveform of the converter will become (EE 2008)

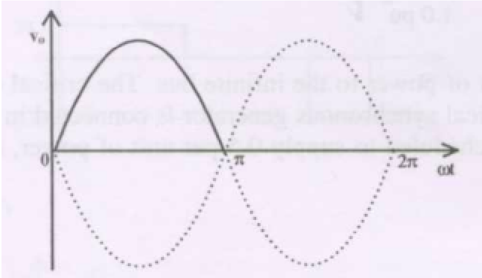


Fig. 1.1.111.2

a)

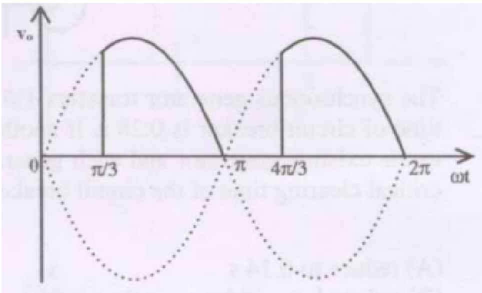


Fig. 1.1.111.3

b)

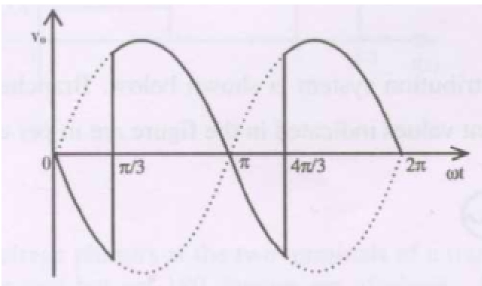


Fig. 1.1.111.4

c)

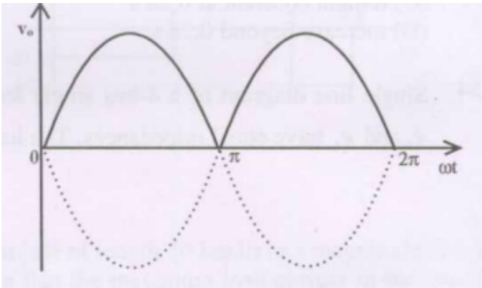


Fig. 1.1.111.5

d)

1.1.112 A single phase voltage source inverter is feeding a purely inductive load as shown in the figure.

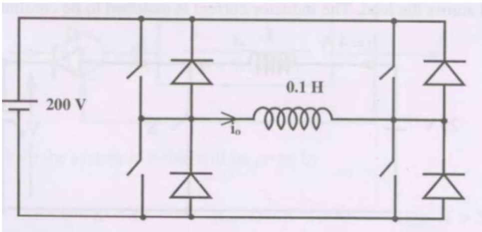


Fig. 1.1.112.1

The inverter is operated at 50 Hz in 180° square wave mode. Assume that the load current does not have any dc component. The peak value of the inductor current i_o will be (EE 2008)

a) 6.37 A b) 10 A c) 20 A d) 40 A

1.1.113 A single phase fully controlled converter bridge is used for electrical braking of a separately excited dc motor. The dc motor load is represented by an equivalent circuit as shown in the figure.

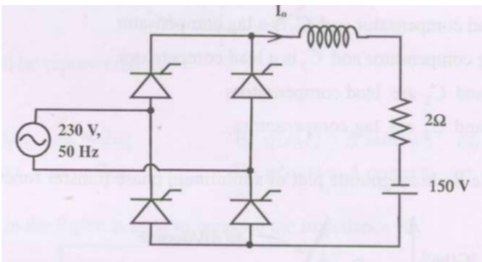


Fig. 1.1.113.1

Assume that the load inductance is sufficient to ensure continuous and ripple free load current. The firing angle of the bridge for a load current of $I_o = 10$ A will be (EE 2008)

a) 44° b) 51° c) 129° d) 136°

1.1.114 A three phase fully controlled bridge converter is feeding a load drawing a constant and ripple free load current of 10 A at a firing angle of 30°. The approximate Total Harmonic Distortion (%THD) and the rms value of fundamental component of the input current will respectively be (EE 2008)

a) 31% and 6.8 A b) 31% and 7.8 A c) 66% and 6.8 A d) 66% and 7.8 A

1.1.115 In the circuit shown in the figure, the switch is operated at a duty cycle of 0.5. A large capacitor is connected across the load. The inductor current is assumed to be continuous. (EE 2008)

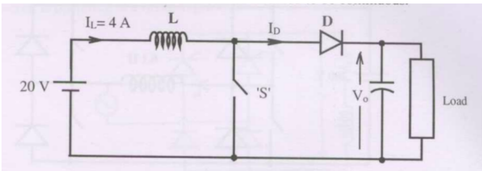


Fig. 1.1.115.1

The average voltage across the load and the average current through the diode will respectively be

a) 10 V, 2 A b) 10 V, 8 A c) 40 V, 2 A d) 40 V, 8 A

1.1.116 The transfer function of a linear time invariant system is given as

$$G(s) = \frac{1}{s^2 + 3s + 2}$$

The steady state value of the output of this system for a unit impulse input applied at time instant $t = 1$ will be (EE 2008)

a) 0 b) 0.5 c) 1 d) 2

1.1.117 The transfer functions of two compensators are given below:

$$C_1 = \frac{10(s+1)}{(s+10)}, \quad C_2 = \frac{s+10}{10(s+1)}$$

Which one of the following statements is correct? (EE 2008)

a) C_1 is a lead compensator and C_2 is a lag compensator
b) C_1 is a lag compensator and C_2 is a lead compensator

c) Both C_1 and C_2 are lead compensators
d) Both C_1 and C_2 are lag compensators

1.1.118 The asymptotic Bode magnitude plot of a minimum phase transfer function is shown in the figure:

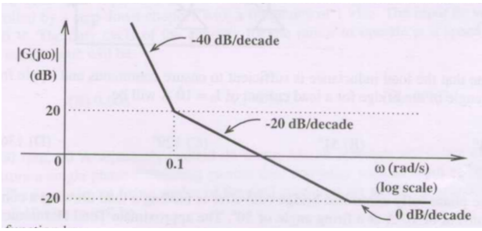


Fig. 1.1.118.1

This transfer function has (EE 2008)

a) Three poles and one zero
b) Two poles and one zero
c) Two poles and two zeros
d) One pole and two zeros

1.1.119 Figure shows a feedback system where $K > 0$.

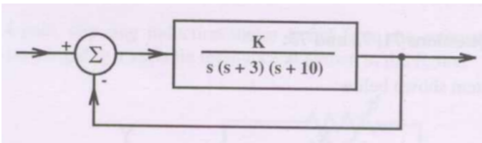


Fig. 1.1.119.1

The range of K for which the system is stable will be given by (EE 2008)

a) $0 < K < 30$
b) $0 < K < 39$
c) $0 < K < 390$
d) $K > 390$

1.1.120 The transfer function of a system is given as (EE 2008)

$$\frac{100}{s^2 + 20s + 100}$$

This system is

a) an overdamped system
b) an underdamped system
c) a critically damped system
d) an unstable system

1.1.121 Two sinusoidal signals $p(\omega_1 t) = A \sin \omega_1 t$ and $q(\omega_2 t)$ are applied to X and Y inputs of a dual channel CRO. The Lissajous figure displayed on the screen is shown below:

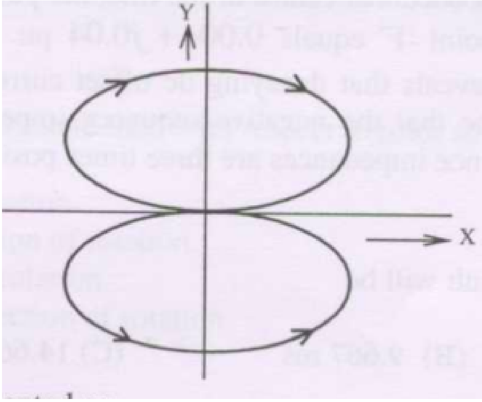


Fig. 1.1.121.1

The signal $q(\omega_2 t)$ will be represented as (EE 2008)

a) $q(\omega_2 t) = A\sin\omega_2 t, \quad \omega_2 = 2\omega_1$
b) $q(\omega_2 t) = A\sin\omega_2 t, \quad \omega_2 = \omega_1/2$
c) $q(\omega_2 t) = A\cos\omega_2 t, \quad \omega_2 = 2\omega_1$
d) $q(\omega_2 t) = A\cos\omega_2 t, \quad \omega_2 = \omega_1/2$

1.1.122 The ac bridge shown in the figure is used to measure the impedance Z.

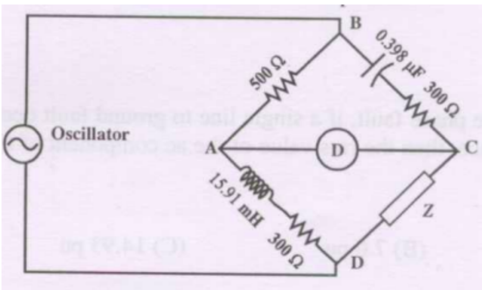


Fig. 1.1.122.1

If the bridge is balanced for oscillator frequency $f = 2$ kHz, then the impedance Z will be (EE 2008)

a) $(260 + j0)\,\Omega$
b) $(0 + j200)\,\Omega$
c) $(260 - j200)\,\Omega$
d) $(260 + j200)\,\Omega$

1.1.123 The current $i(t)$ sketched in the figure flows through an initially uncharged 0.3 nF capacitor. (EE 2008)

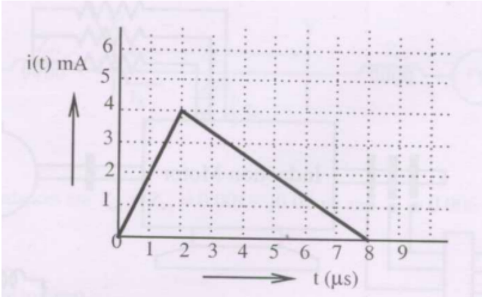


Fig. 1.1.123.1

a) The charge stored in the capacitor at $t = 5\ \mu$s will be

i) 8 nC ii) 10 nC iii) 13 nC iv) 16 nC

b) The capacitor charged up to 5 μs, as per the current profile given in the figure, is connected across an inductor of 0.6 mH. Then the value of voltage across the capacitor after 1 μs will approximately be

i) 18.8 V ii) 23.5 V iii) −23.5 V iv) −30.6 V

1.1.124 The state space equation of a system is described by (EE 2008)

$$\mathbf{x} = \mathbf{Ax} + \mathbf{Bu}$$
$$\mathbf{y} = \mathbf{Cx}$$

where **x** is state vector, **u** is input, **y** is output and

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & -2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \mathbf{C} = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

a) The transfer function $G(s)$ of this system will be

i) $\frac{s}{(s+2)}$ ii) $\frac{s+1}{s(s-2)}$ iii) $\frac{s}{(s-2)}$ iv) $\frac{1}{s(s+2)}$

b) A unity feedback is provided to the above system $G(s)$ to make it a closed-loop system as shown.

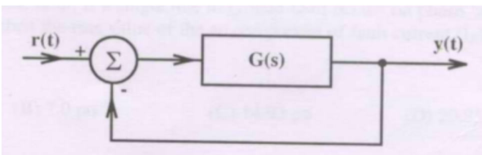


Fig. 1.1.124.1

For a unit step input $r(t)$, the steady state error in the output will be (EE 2008)

i) 0 ii) 1 iii) 2 iv) ∞

1.1.125 A general filter circuit is shown in the figure:

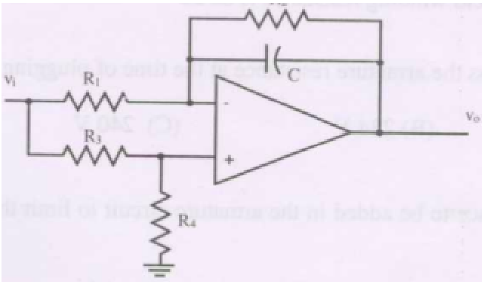


Fig. 1.1.125.1

If $R_1 = R_2 = R_A$ and $R_3 = R_4 = R_B$, the circuit acts as a (EE 2008)

a) all pass filter
b) band pass filter
c) high pass filter
d) low pass filter

1.1.126 The output of the filter in Fig. 1.1.125.1 is given to the following circuit shown

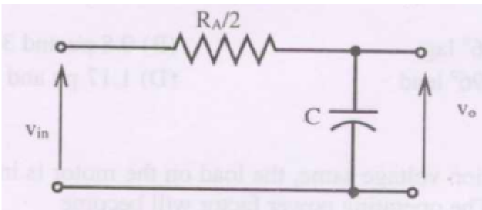


Fig. 1.1.126.1

The gain vs frequency characteristic of the output (v_o) will be (EE 2008)

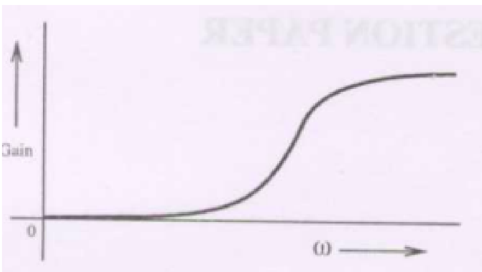


Fig. 1.1.126.2

a)

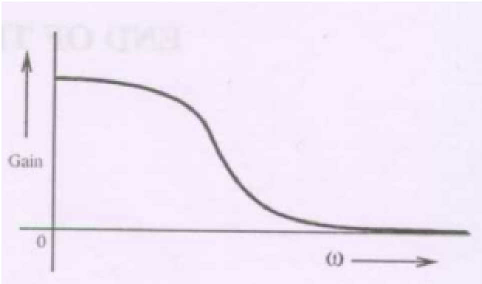


Fig. 1.1.126.3

b)

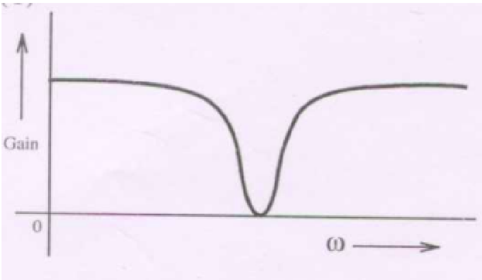


Fig. 1.1.126.4

c)

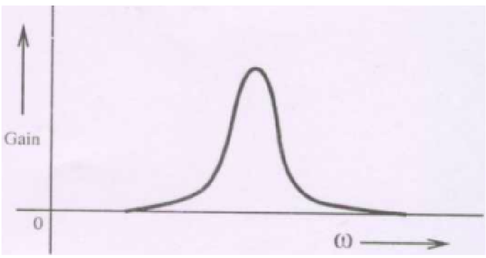


Fig. 1.1.126.5

d)

1.1.127 If a curent of $\left[-6\sqrt{2}\sin(100\pi t) + 6\sqrt{2}\cos\left(300\pi t + \frac{\pi}{4}\right) + 6\sqrt{2}\right] A$ is passed through a true RMS ammeter , the meter reading will be (IN 2008)

a) $6\sqrt{2}$ A b) $\sqrt{126}$ A c) 12 A d) $\sqrt{216}$ A

1.1.128 If the ac bridge circuit shown below Fig. 1.1.128.1 is balanced, the element Z can be a (IN 2008)

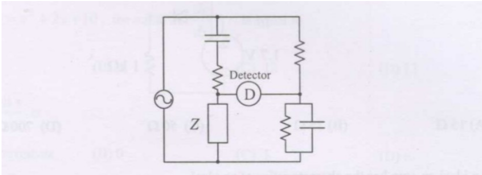


Fig. 1.1.128.1

a) pure capacitor
b) pure inductor
c) R-L series combination
d) R-L parallel combination

1.1.129 The Bode asymptotic plot of a transfer function is given belowFig. 1.1.129.1. In the frequency range shown, the transfer function has (IN 2008)

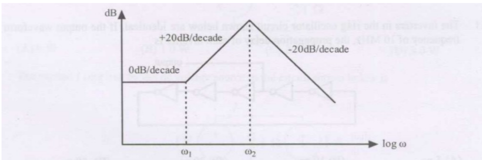


Fig. 1.1.129.1

a) 3 poles and 1 zero
b) 1 pole and 2 zeros
c) 2 poles and 1 zero
d) 2 poles and 2 zeros

1.1.130 The Fourier transform of $x(t) = e^{-at}u(-t)$, where $u(t)$ is the unit step function, (IN 2008)

a) exists for any real value of a
b) does not exist for any real value of a
c) exists if the real value of a is strictly negative
d) exists if the real value of a is strictly positive

1.1.131 In the circuit shown below in Fig. 1.1.131.1, the maximum power that can be transferred to the load ZL is (IN 2008)

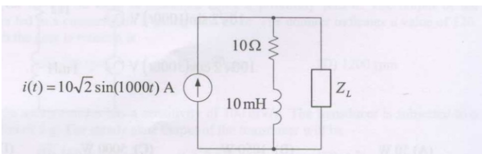


Fig. 1.1.131.1

a) 250 W b) 500 W c) 1000 W d) 2000 W

1.1.132 In the circuit shown below in Fig. 1.1.132.1 the average power consumed by the 1Ω resistor is (IN 2008)

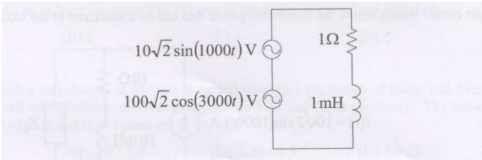


Fig. 1.1.132.1

a) 50 W b) 1050 W c) 5000 W d) 10100 W

1.1.133 For the circuit shown below Fig. 1.1.133.1 the steady-state current I is (IN 2008)

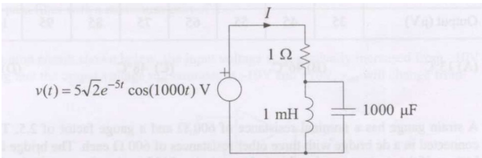


Fig. 1.1.133.1

a) 0 A
b) $5\sqrt{2}\cos(1000t)\,A$
c) $5\sqrt{2}\cos\left(1000t - \frac{\pi}{4}\right)A$
d) $5\sqrt{2}A$

1.1.134 For the circuit shown below Fig. 1.1.134.1 the voltage across the capacitor is (IN 2008)

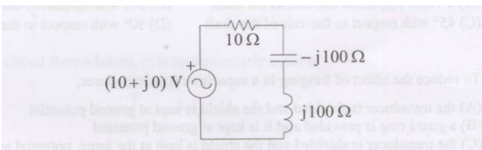


Fig. 1.1.134.1

a) $(10 + j0)$ V b) $(100 + j0)$ V c) $(0 + j100)$ V d) $(0 - j100)$ V

1.1.135 The op-amp circuit shown below Fig. 1.1.135.1 is that of a (IN 2008)

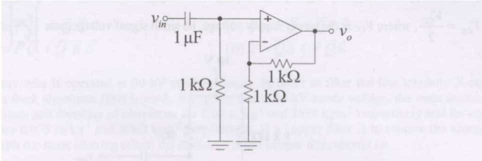


Fig. 1.1.135.1

a) low-pass filter with a maximum gain of 1
b) low-pass filter with a maximum gain of 2
c) high-pass filter with a maximum gain of 1
d) high-pass filter with a maximum gain of 2

1.1.136 If the bandwidth of a low-pass signal g(t) is 3 kHz, the bandwidth of $g^2(t)$ will be (IN 2008)

a) $\frac{3}{2}$ MHz   b) 3 MHz   c) 6 kHz   d) 9 kHz

1.1.137 Consider the AM signal $s(t) = [1 + m(t)] \cos 2\pi f t$. It is given that the bandwidth of the real, low-pass message signal $m(t)$ is 2 kHz. If f_c = 2 MHz, the bandwidth of the band-pass signal $s(t)$ will be (IN 2008)

a) 2.004 Hz   b) 2 kHz   c) 4 kHz   d) 2 kHz

1.1.138 Ten real, band-pass message signals, each of bandwidth 3 kHz, are to be frequency division multiplexed over a band-pass channel with bandwidth B kHz. If the guard band in between any two adjacent signals should be of 500 Hz width and there is no need to provide any guard band at the edges of the band-pass channel, the value of B should be at least (IN 2008)

a) 30   b) 34.5   c) 35   d) 35.5

1.1.139 The step response of a linear time invariant system is $y(t)$ = 5 e^{-10t} $u(t)$, where $u(t)$ is the unit step function. If the output of the system corresponding to an impulse input $\delta(t)$ is $h(t)$, then $h(t)$ is (IN 2008)

a) -50 e^{-10t} $u(t)$
b) 5 e^{-10t} $\delta(t)$
c) 5 $u(t)$ -50 e^{-10t} $\delta(t)$
d) 5 $\delta(t)$ -50 e^{-10t} $u(t)$

1.1.140 The op-amp based circuit of a half wave rectifier electronic voltmeter shown below Fig. 1.1.140.1 uses a $PMMC$ ammeter with a full scale deflection (FSD) current of 1 mA and a coil resistance of 1 kΩ. The value of R that gives FSD for a sinusoidal input voltage of 100 mV (RMS) is (IN 2008)

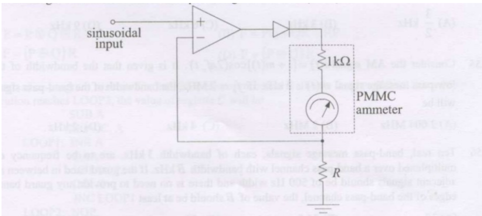


Fig. 1.1.140.1

a) 45 Ω b) 67.5 Ω c) 100 Ω d) 144.4 Ω

1.1.141 The x and y sensitivities of an analog oscilloscope are set as 2 ms/cm and 1V/cm respectively. The trigger is set at $0V$ with negative slope. An input of 2 $\cos(100\pi t + 30°)$ V is fed to the y input of the oscilloscope. The waveform seen on the oscilloscope will be (IN 2008)

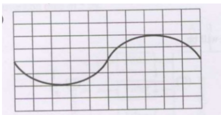


Fig. 1.1.141.1

a)

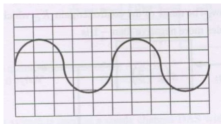


Fig. 1.1.141.3

c)

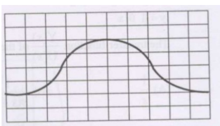


Fig. 1.1.141.2

b)

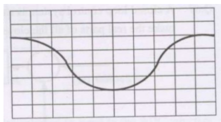


Fig. 1.1.141.4

d)

1.1.142 The open loop transfer function of a unity feedback system is $G(s) = \frac{K(s+2)}{(s+1+j1)(s+1-j1)}$. The root locus plot of the system has (IN 2008)

a) two breakaway points located at s = -0.59 and s = -3.41
b) one breakaway point located at s = -0.59
c) one breakaway point located at s =-3.41
d) one breakaway point located at s =-1.41

1.1.143 If a first order system and its time response to a unit step input are as shown below in Fig. 1.1.143.1, the gain K is (IN 2008)

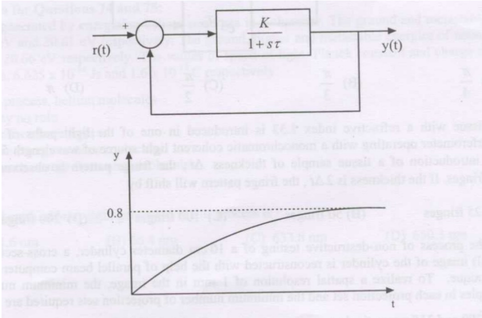


Fig. 1.1.143.1

a) 0.25 b) 0.8 c) 1 d) 4

1.1.144 The state space representation of a system is given by

$$\dot{x} = \begin{pmatrix} 0 & 1 \\ 0 & -3 \end{pmatrix} x + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u$$
$$y = \begin{pmatrix} 1 & 0 \end{pmatrix} x$$

The transfer function $\frac{Y(s)}{U(s)}$ of the system will be (IN 2008)

a) $\frac{1}{s}$ b) $\frac{1}{s(s+3)}$ c) $\frac{1}{s+3}$ d) $\frac{1}{s^2}$

1.1.145 A closed loop control system is shown below in Fig. 1.1.145.1. The range of the controller gain K_C which will make the real parts of all the closed loop poles more negative than -1 is (IN 2008)

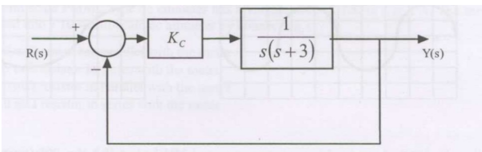


Fig. 1.1.145.1

a) K_C > -4 b) K_C > 0 c) K_C > 2 d) K_C < 2

1.1.146 For the closed loop system shown below Fig. 1.1.146.1 to be stable, the value of time delay T_D (in seconds) should be less than (IN 2008)

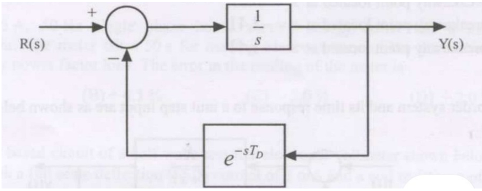


Fig. 1.1.146.1

a) $\frac{\pi}{4}$ b) $\frac{\pi}{3}$ c) $\frac{\pi}{2}$ d) π

1.1.147 In the circuit shown below Fig. 1.1.147.1 the steady-state is reached with the switch K open. Subsequently the switch is closed at time t = 0.

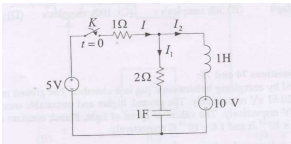


Fig. 1.1.147.1

a) At time t = 0^+, current I is (IN 2008)

i) -$\frac{5}{3}$ A ii) 0 A iii) $\frac{5}{3}$ A iv) ∞ A

b) At time t = 0^+, $\frac{dI_2}{dt}$ (IN 2008)

i) -5 A/s ii) -$\frac{10}{3}$ A/s iii) 0 A/s iv) 5A/s

1.1.148 Consider a unity feedback system with open loop transfer function $G(s) = \frac{1+6s}{s^2(1+s)(1+2s)}$ (IN 2008)

a) The phase crossover frequency of the system in radians per second is

i) 0.125 ii) 0.25 iii) 0.5 iv) 1

b) The gain margin of the system is

i) 0.125 ii) 0.25 iii) 0.5 iv) 1

1.1.149 A unity feedback system has open loop transfer function $G(s) = \frac{100}{s(s+p)}$. The time at which the response to a unit step input reaches its peak is $\frac{\pi}{8}$ seconds. (IN 2008)

a) The damping coefficient for the closed loop system is

i) 0.4 ii) 0.6 iii) 0.8 iv) 1

b) The value of p is

i) 6 ii) 12 iii) 14 iv) 16

1.2 Discrete

1.2.1 The Euler iteration formula for numerically integrating a first order nonlinear differential equation of form $x = f(x)$, with a constant step size of Δt is (AE 2007)

a) $x_{k+1} = x_k - \Delta t \times f(x_k)$
b) $x_{k+1} = x_k + \left(\Delta t^2/2\right) \times f(x_k)$
c) $x_{k+1} = x_k - (1/\Delta t) \times f(x_k)$
d) $x_{k+1} = x_k + \Delta t \times f(x_k)$

1.2.2 Numerical values of the integral $\int_0^1 \frac{1}{1+x^2}\,dx$ if evaluated numerically using the Trapazoidal rule with dx = 0.2 would be (AE 2007)

a) 1 b) $\pi/4$ c) 0.7837 d) 0.2536

1.2.3 The Newton-Raphson iteration formula to find a cube root of a positive number c is (AE 2007)

a) $x_{k+1} = \frac{2x_k^3 + \sqrt[3]{c}}{3x_k^2}$ b) $x_{k+1} = \frac{2x_k^3 - \sqrt[3]{c}}{-3x_k^2}$ c) $x_{k+1} = \frac{2x_k^3 + c}{3x_k^2}$ d) $x_{k+1} = \frac{x_k^3 + c}{3x_k^2}$

1.2.4 $y(0) = 0$ and using Euler's method with step size $h = 0.1$ solution of the differential equation

$$\frac{dy}{dx} = 2xy + 1$$

gives the value of $y(0.3)$ =? (AG 2007)

a) 0.3101 b) 0.3142 c) 0.6202 d) 4.080

1.2.5 Integrating the function (AG 2007)

$$f(x) = 1 + e^{-x}\sin(4x)$$

over the interval [0, 1] using Simpson's $\frac{1}{3}$ rule gives Result = ?

a) 1.021 b) 1.091 c) 1.321 d) 2.642

1.2.6 The degree of the differential equation $\frac{d^2x}{dt^2} + 2x^3 = 0$ is (CE 2007)

a) 0 b) 1 c) 2 d) 3

1.2.7 The solution of the differential equation $\frac{dy}{dx} = x^2y$ with the condition that $y = 1$ at $x = 0$ is (CE 2007)

a) $y = e^{\frac{1}{2x}}$ b) $\ln(y) = \frac{x^3}{3} + 4$ c) $\ln(y) = \frac{x^2}{2}$ d) $y = e^{\frac{x^3}{3}}$

1.2.8 The following equation needs to be numerically solved using the Newton-Raphson method. $x^3 + 4x - 9 = 0$ The iterative equation for this purpose is (k indicates the iteration level) (CE 2007)

a) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+4}$
b) $x_{k+1} = \frac{2x_k^3+9}{3x_k^2+9}$
c) $x_{k+1} = x_k - \frac{x_k^3+4x_k-9}{3x_k^2+4}$
d) $x_{k+1} = \frac{4x_k^2+3}{9x_k^2+2}$

1.2.9 Given that one root of the equation $x^3 - 10x^2 + 31x - 30 = 0$ is 5, the other two roots are (CE 2007)

a) 2 and 3 b) 2 and 4 c) 3 and 4 d) -2 and -3

1.2.10 Consider the series $x_{n+1} = \frac{x_n}{2} + \frac{9}{8x_n}, x_0 = 0.5$ obtained from the Newton-Raphson method. The series converges to (CS 2007)

a) 1.5 b) $\sqrt{2}$ c) 1.6 d) 1.4

1.2.11 The following plot shows a function y which varies linearly with x. The value of the integral $\int_0^2 y\,dx$ is: (EC 2007)

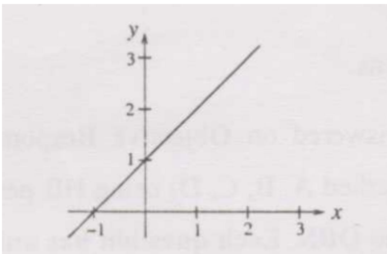


Fig. 1.2.11.1

a) 1.0 b) 2.5 c) 4.0 d) 5.0

1.2.12 For $|x| < 1$, $\coth(x)$ can be approximated as: (EC 2007)

a) x b) x^2 c) $\frac{1}{x}$ d) $\frac{1}{x^2}$

1.2.13 $\lim_{\theta \to 0} \frac{\sin(\theta/2)}{\theta}$ is: (EC 2007)

a) 0.5 b) 1 c) 2 d) Not defined

1.2.14 Which of the following functions is strictly bounded? (EC 2007)

a) $\sin x$ b) e^x c) $\cos x$ d) x^2

1.2.15 For the function e^{-x}, the linear approximation around $x = 2$ is: (EC 2007)

a) $(3 - x)^2$
b) $1 - x$
c) $[3 + 2\sqrt{2} - (1 + \sqrt{2})x]e^2$
d) e^{-2}

1.2.16 The equation $x^2 + x^4 - 4x - 4 = 0$ is to be solved using Newton-Raphson. If $x = 2$ is the initial approximation, then the next approximation using this method will be: (EC 2007)

a) $\frac{2}{3}$ b) $\frac{4}{3}$ c) 1 d) $\frac{3}{2}$

1.2.17 A 5-point sequence $x[n]$ is given as $x[-3] = 1$, $x[-2] = 1$, $x[-1] = 0, x[0] = 5$, $x[1] = 15$. Let $X\left(e^{j\omega}\right)$ denote the discrete-time Fourier transform of $x[n]$. The value of $\int_{-\pi}^{\pi} X\left(e^{4j\omega 3}\right) d\omega$ (EC 2007)

a) 5 b) 10π c) 16π d) $5 + j10\pi$

1.2.18 The z-transform $X[z]$ of a sequence $x[n]$ is given by $X[z] = \frac{0.5}{1-2z^{-1}}$. It is given that the region of convergence of $X[z]$ includes the unit circle. The value of $x[0]$ is: (EC 2007)

a) -0.5 b) 0 c) 0.25 d) 0.5

1.2.19 The differential equation $\frac{dx}{dy} = \frac{1-x}{\tau}$ is discretised using Euler's numerical integration method with a time step triangle $\Delta T > 0$. What is the maximum permissible value of ΔT to ensure stability of the solution of the corresponding discrete time equation? (EE 2007)

a) 1 b) $\frac{\tau}{2}$ c) τ d) 2τ

1.2.20 Consider the discrete-time system shown in the figure where the impulse response $G(z)$ is $g(0) = 0$, $g(1) = g(2) = 1$, $g(3) = g(4) = \cdots = 0$. (EE 2007)

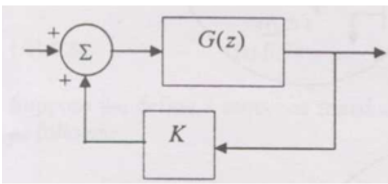


Fig. 1.2.20.1

This system is stable for range of values of K

a) $[-1, 1/2]$ b) $[-1, 1]$= c) $[-1/2, 1]$ d) $[-1/2, 2]$

1.2.21 The integral $\frac{1}{2\pi}\int_0^{2\pi} \sin(t-\tau)\cos\tau\, d\tau$ equals (EE 2007)

a) $\sin t \cos t$ b) 0 c) $(1/2)\cos t$ d) $(1/2)\sin t$

1.2.22 $X(z) = 1-3z^{-1}, Y(z) = 1+2z^{-2}$ are Z-transforms of two signals $x[n]$, $y[n]$ respectively. A linear time invariant system has the impulse response $h[n]$ defined by these two signals as $h[n] = x[n-1] * y[n]$ where $*$ denotes discrete time convolution. Then the output of the system for the input $\delta[n-1]$ (EE 2007)

a) has Z-transform $z^{-1}X(z)Y(z)$
b) equals $\delta[n-2] - 3\delta[n-3] + 2\delta[n-4] - 6\delta[n-5]$
c) has Z-transform $1 - 3z^{-1} + 2z^{-2} - 6z^{-3}$
d) does not satisfy any of the above three.

1.2.23 The minimum frequency at which a signal comprising of 30 Hz, 50 Hz and 70 Hz should be sampled to avoid aliasing is ____________ Hz. (GE 2007)

1.2.24 Convolving two sampled signals $f(n) = \{1, 1, 2, 2\}$ with $g(n) = \{3, 2, 1\}$ results in a function $x(n)$ equal to (GG 2007)

a) {1, 3, 7, 9, 10, 6}
b) {3, 5, 9, 11, 6, 2}
c) {3, 9, 6, 11, 2, 2}
d) {3, 5, 9, 6, 11, 5}

1.2.25 Let $x(t)$ be a continuous-time, real-valued signal band-limited to F Hz. The Nyquist sampling rate, in Hz, for $y(t) = x(0.5t) + x(t) - x(2t)$ is (IN 2007)

a) F b) $2F$ c) $4F$ d) $8F$

1.2.26 Consider the periodic signal $x(t) = (1+0.5\cos 40\pi t)\cos 200\pi t$, where t is in seconds. Its fundamental frequency, in Hz, is (IN 2007)

a) 20 b) 40 c) 100 d) 200

1.2.27 The polynomial $p(x) = x^5 + x^2 + 2$ has (IN 2007)

a) all real roots
b) 3 real and 2 complex roots
c) 1 real and 4 complex roots
d) all complex roots

1.2.28 Identify the Newton-Raphson iteration scheme for finding the square root of 2. (IN 2007)

a) $x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$
b) $x_{n+1} = \frac{1}{2}\left(x_n - \frac{2}{x_n}\right)$
c) $x_{n+1} = \frac{2}{x_n}$
d) $x_{n+1} = \sqrt{2}\, x_n$

1.2.29 Consider the discrete-time signal $x(n) = (-1)^n u(n)$, where $u(n) = 1$ for $n \geq 0$, and define $y(n) = x(-n)$. Then $\sum_{n=-\infty}^{\infty} y(n)$ equals: (IN 2007)

a) $-2/3$ b) $2/3$ c) $3/2$ d) 3

1.2.30 Suppose $y_p(x) = x\cos(2x)$ is a particular solution of

$$y'' + \alpha y = \sin(2x).$$

Then the constant α equals (MA 2007)

a) -4 b) -2 c) 2 d) 4

1.2.31 The smallest degree of the polynomial that interpolates the data

x	-2	-1	0	1	2	3
$f(x)$	-58	-21	-12	-13	-6	27

TABLE 1.2.31

is: (MA 2007)

a) 3 b) 4 c) 5 d) 6

1.2.32 Suppose that x_n is sufficiently close to 3. Which of the following iterations $x_{n+1} = g(x_n)$ will converge to the fixed point $x = 3$? (MA 2007)

a) $x_{n+1} = -16 + 6x_n + \dfrac{3}{x_n}$

b) $x_{n+1} = \sqrt{3 + 2x_n}$

c) $x_{n+1} = \dfrac{3}{x_n} - \dfrac{x_n}{2}$

d) $x_{n+1} = \dfrac{x_n^2 - 3}{2}$

1.2.33 Consider the quadrature formula:

$$\int_{x_1}^{x_2} f(x)\,dx \approx \frac{1}{2}\left[f(x_1) + f(x_2)\right],$$

where x_1 and x_2 are quadrature points. Then the highest degree of the polynomial for which the above formula is exact equals: (MA 2007)

a) 1 b) 2 c) 3 d) 4

1.2.34 The value of

$$\int_0^1 \int_y^1 x^2 e^{x^2}\,dx\,dy$$

equals (MA 2007)

a) $\frac{1}{4}$ b) $\frac{1}{3}$ c) $\frac{1}{2}$ d) 1

1.2.35

$$\lim_{n\to\infty}\left[(n+1)\int_0^1 x^n \ln(1+x)\,dx\right] =$$

(MA 2007)

a) 0 b) $\ln 2$ c) $\ln 3$ d) ∞

1.2.36 Which of the following inequality is NOT true for $x \in \left[\frac{1}{4}, \frac{3}{4}\right]$ (MA 2007)

a) $e^{-x} > \sum_{j=0}^{\infty} \frac{(-x)^j}{j!}$
b) $e^{-x} < \sum_{j=0}^{\infty} \frac{(-x)^j}{j!}$
c) $e^{-x} = \sum_{j=0}^{\infty} \frac{(-x)^j}{j!}$
d) $e^{-x} > \sum_{j=0}^{10} \frac{(-x)^j}{j!}$

1.2.37 If $y = x + \sqrt{x + \sqrt{x + \sqrt{x + \cdots}}}$, then $y(2) =$ (ME 2007)

a) 4 or 1 b) 4 only c) 1 only d) undefined

1.2.38 The solution of $\frac{dy}{dx} = y^2$ with initial value $y(0) = 1$ is bounded in the interval (ME 2007)

a) $-\infty < x < \infty$ b) $-\infty < x < 1$ c) $x < 1, x > 1$ d) $-2 \le x \le 2$

1.2.39 A calculator has accuracy up to 8 digits after decimal place. The value of $\int_0^1 \sin x\,dx$ when evaluated using this calculator by trapezoidal method with 8 equal intervals, to 5 significant digits is (ME 2007)

a) 0.00000 b) 1.0000 c) 0.00500 d) 0.00025

1.2.40 $\lim_{x\to 0} \frac{e^x - \left(1+x+\frac{x^2}{2}\right)}{x^3} =$ (ME 2007)

a) 0 b) $\frac{1}{6}$ c) $\frac{1}{3}$ d) 1

1.2.41 The solution of $ye^x dx + (4y + e^x)dy = 0$ for $y(0) = -1$ is (MN 2007)

a) $ye^x + 2y^2 - 1 = 0$
b) $e^x + y^2 - 2 = 0$
c) $ye^x - y^2 = 0$
d) $ye^x + y^2 - 1 = 0$

1.2.42 Consider the following data for the grade of iron ore from a working bench over past 5 weeks:

Week	Grade (% Fe)
1	62.1
2	61.0
3	60.5
4	62.5
5	62.0

TABLE 1.2.42

The 3-week moving average forecast for the grade, in % Fe, in the 6th week is:
(MN 2007)

a) 61.66 b) 61.90 c) 62.20 d) 62.50

1.2.43 The values of $f(x)$ at x_0, x_1 and x_2 are 9.0, 12.0 and 15.0 respectively. Using Simpson's $\frac{1}{3}$ rule, the value of $\int f(x)\,dx$, considering an interval of 0.1 is:
(MN 2007)

a) 1.2 b) 2.4 c) 1.6 d) 1.8

1.2.44 Solution of the differential equation $x\frac{dy}{dx} + y = x^4$, with the boundary condition that $y = 1$ at $x = 1$, is
(PH 2007)

a) $y = 5x^4 - 4$ b) $y = x^4 + \frac{4x}{5}$ c) $y = \frac{4x^4}{5} + \frac{1}{5x}$ d) $y = x^4 + \frac{4}{5x}$

1.2.45 What is the value of

$$\lim_{x \to \frac{\pi}{4}} \frac{\cos x - \sin x}{x - \frac{\pi}{4}}$$

Fig. 1.2.45.1

(2007 PI)

a) $\sqrt{2}$
b) 0
c) $-\sqrt{2}$
d) Limit does not exist

1.2.46 The function e^x over the interval [0,1] is to be evaluated using the Taylor series $1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots$ to accuracy of $\delta > 0$.The number of terms in the series that is considered for this accuracy is n. Then
(2007 PI)

a) for a given $x \in [0, 1]$ and a given δ, there is no finite n that is valid
b) for a given $\delta > 0$, there is a valid n that is finite for a given $x \in [0, 1]$, but there is no finite n that is valid for all $x \in [0, 1]$

c) for a given $\delta > 0$,there is a finite n that is valid for all $x \in [0, 1]$
d) there is a finite n that is valid for all x in [0,1] an all $\delta > 0$

1.2.47 $\lim_{x \to \frac{\pi}{2}} \frac{\sin^2(2x)}{\left(x - \frac{\pi}{2}\right)^2}$ is equal to : (2007 XE)

a) -4 b) 0 c) 2 d) 4

1.2.48 The equation g(x) = x is solved by Newton-Raphson iteration method, starting with an initial approximation x_n near the simple root a. If x_{n+1} is the approximation to a at the (n+1)th iteration, then: (2007 XE)

a) $x_{n+1} = \frac{x_n g'(x_n) - g(x_n)}{1 - g'(x_n)}$
b) $x_{n+1} = \frac{x_n, g'(x_n) - g(x_n)}{g'(x_n) - 1}$
c) $x_{n+1} = g(x_n)$
d) $x_{n+1} = \frac{x_n, g'(x_n) - g(x_n) + 2x_n}{g'(x_n) + 1}$

1.2.49 If Runge–Kutta method of order 4 is used to solve the differential equation $\frac{dy}{dx} = f(x)$, y(0) = 0, in the interval [0,h] with step size h, then (2007 XE)

a) $y(h) = \frac{h}{6}[f(0) + 4f(h/2) + f(h)]$
b) $y(h) = \frac{h}{6}[f(0) + f(h)]$
c) $y(h) = h[f(0) + f(h)]$
d) $y(h) = \frac{h}{6}[f(0) + 2f(h/2) + f(h)]$

1.2.50 If a polynomial of degree three interpolates a function f(x) at the points (0,3), (1,13), (3,99), (4,187), then f(2) is: (2007 XE)

a) 20 b) 36 c) 43 d) 58

1.2.51 Let $f(x) = x^2$ for $-\pi \leq x \leq \pi$ and f(x+2π) = f(x).

a) The Fourier series of f in $[-\pi, \pi]$ is: (2007 XE)

i) $\frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$
ii) $\frac{\pi^2}{3} + 4(-1)^n \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$
iii) $\frac{\pi^2}{3} + 4(-1)^n \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$
iv) $\frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$

b) The sum of the absolute values of the Fourier coefficients of f is: (2007 XE)

i) $\frac{\pi^2}{6}$ ii) $\frac{\pi^2}{3}$ iii) $\frac{2\pi^2}{3}$ iv) π^2

1.2.52 Let $y(x) = \sum_{n=0}^{\infty} a_n x^n$ be a solution of $\frac{d^2y}{dx^2} + xy = 0$. (2007 XE)

a) The value of a_n is:

i) 0 ii) 1 iii) 2 iv) 3

b) The solution of the above differential equation satisfying y(0) = 1 and y'(0) = 0 is:

i) $y(x) = 1 + \frac{x^2}{2!\cdot 3} - \frac{x^4}{2\cdot 3\cdot 5\cdot 6} + \ldots$ iii) $y(x) = 1 + \ldots$
ii) $y(x) = 1 - \frac{2\cdot 3x^2}{1} + \ldots$ iv) $y(x) = 1 - \ldots$

1.2.53 If a root of $f(x) = x^2 - 2x + 1 = 0$ is obtained by using iteration $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$ with $x_0 = 0.5$, the convergence rate is
(2007 XE)

a) 1 b) 1.62 c) 1.84 d) 2

1.2.54 If y_i denotes the value of y(x) at x = x_i in $x_0 < x_1 < \cdots < x_n$, $x_i - x_{i-1} = h$ for $1 \le i \le n$, then $\frac{d^2y}{dx^2}$ at $x = x_i$ is approximated using finite differences by:
(2007 XE)

a) $\frac{y_{i+1}-2y_i+y_{i-1}}{h^2}$ b) $\frac{y_{i+1}-y_i+y_{i-1}}{h^2}$ c) $\frac{y_{i+1}-2y_i+y_{i-1}}{2h}$ d) $\frac{y_{i+1}-y_i+y_{i-1}}{2h^2}$

1.2.55 The minimum number of terms required in e^x series expansion to evaluate at x = 1 correct up to 3 decimal places is: (2007 XE)

a) 8 b) 7 c) 6 d) 5

1.2.56 The iteration $x_{n+1} = \frac{1}{1+x_n^2}$ converges to a real number x in the interval (0,1) with $x_0 = 0.5$. The value of x correct up to 2 decimal places is: (2007 XE)

a) 0.65 b) 0.68 c) 0.73 d) 0.80

1.2.57 The graph y = f(x) passes through (0,-3), (1,-1), (2,3). Using Lagrange interpolation, the x-value where y = 0 is: (2007 XE)

a) 1.375 b) 1.475 c) 1.575 d) 1.675

1.2.58 The equation of the best fit line (least squares) for: $x : 1\ 2\ 3\ 4\ 5$, $y : 14\ 13\ 9\ 5\ 2$ is: (2007 XE)

a) y = 18 - 3x b) y = 18.1 - 3.1x c) y = 18.2 - 3.2x d) y = 18.3 - 3.3x

1.2.59 Solve $\frac{dy}{dx} = xy^2$, y(1) = 1 by Euler's method, step h = 0.1, find y(1.2): (2007 XE)

a) 1.1000 b) 1.1232 c) 1.2210 d) 1.2331

1.2.60 The local error of scheme: $y_{n+1} = y_n + \frac{h}{12}(5y_{n+1} + 8y_n - y_{n-1})$ by comparison with Taylor series is: (2007 XE)

a) O(h) b) $O(h^2)$ c) $O(h^3)$ d) $O(h^4)$

1.2.61 The area between $y = 1 - x^2$ and x-axis from x = -1 to x = 1 using Trapezoidal rule with h = 0.5 is:
(2007 XE)

a) 1.20 b) 1.23 c) 1.25 d) 1.33

1.2.62 Iteration: $x_{n+1} = \sqrt{a}\left(1 + \frac{1}{3a^2}\right)$, $a > 0$ converges to: (2007 XE)

a) $\sqrt{a}$ b) a c) $\sqrt[3]{a}$ d) a^2

1.2.63 Quadrature formula: $\int_0^1 f(x)\,dx \approx \frac{1}{8}\left[f(0) + 2bf(0.25) + 2f(0.5) + 2df(0.75) + f(1)\right]$. If used as Simpson's $\frac{1}{3}$ rule: (2007 XE)

a) b = d = 1 b) b = d = 2 c) b = 2d = 1 d) b = 2d = 2

1.2.64 Using b, d from Q25, $\int_0^1 \frac{dx}{1+x}$ correct to 4 decimal places is: (2007 XE)

a) 0.3091 b) 0.3121 c) 0.3151 d) 0.3191

1.2.65 Solve $\frac{dy}{dx} = f(x, y) = 2xy$, y(0) = 1, y(0.2) = 1.0408 , y(0.4) = 1.1735, y(0.6) = 1.4333. Predictor scheme: (2007 XE)

a) $y_{n+1} = y_n + \frac{4h}{3}(2f_{n-1} - f_{n-2} + 2f_{n-3})$
c) $y_{n+1} = y_{n-1} + \frac{h}{3}(4f_n - 5f_{n-1} + 4f_{n+1})$
b) $y_{n+1} = y_{n-3} + 3\frac{h}{4}(2f_{n-2} - f_{n-1} + 2f_n)$
d) $y_{n+1} = y_{n-3} + \frac{h}{4}(2f_{n-1} - f_{n-2} + 2f_{n-3})$

1.2.66 Using correct predictor from Q27, y(0.8) =: (2007 XE)

a) 1.8680 b) 1.8750 c) 1.8890 d) 1.9055

1.2.67 Which of the following equations is a LINEAR ordinary differential equation?
(AE 2008)

a) $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y^2 = 0$
c) $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 2y = 0$
b) $\frac{d^2y}{dx^2} + y\frac{dy}{dx} + 2y = 0$
d) $\left(\frac{dy}{dx}\right)^2 + \frac{dy}{dx} + 2y = 0$

1.2.68 $\int_0^{\pi/2} \frac{\cos\theta}{\sqrt{1-\cos^2\theta}}d\theta$ is (AG 2008)

a) 0 b) $\frac{\pi}{2}$ c) 1 d) π

1.2.69 A function $f(x)$ is evaluated as 1, 1.5, 2.2 and 3.4 at four values of x having intervals of 0.5. The area under the curve $f(x)$ using trapezoidal rule is (AG 2008)

a) 1.95 b) 2.45 c) 2.95 d) 3.45

1.2.70 Solution of the ordinary differential equation

$$\frac{dy}{dx} = x^2 + 2y$$

is (AG 2008)

a) $y = \frac{2}{3}x^2 + 4x$
b) $y = \sqrt{\frac{2}{3}x^2 + 4x + k}$
c) $y = \frac{2}{3}x^3 - 4x + k$
d) $y = \frac{2}{3}x^3 + 4x + k$

1.2.71 The general solution of

$$\frac{d^2y}{dx^2} + y = 0$$

is (CE 2008)

a) $y = P\cos x + Q\sin x$
b) $y = P\cos x$
c) $y = P\sin x$
d) $y = P\sin^2 x$

1.2.72 The value of

$$\int_0^3 \int_0^{3-x} (6 - x - y)\, dx\, dy$$

is (CE 2008)

a) 13.5
b) 27.0
c) 40.5
d) 54.0

1.2.73 Solution of

$$\frac{dy}{dx} = \frac{-x}{y}$$

at $x = 1$ and $y = \sqrt{3}$ is (CE 2008)

a) $x^2 - y^2 = -2$ b) $x^2 + y^2 = 4$ c) $x^2 - y^2 = -2$ d) $x^2 + y^2 = 4$

1.2.74 Which ONE of the following is NOT an integrating factor for the differential equation $xdy - ydx = 0$? (CH 2008)

a) $\frac{1}{x^2}$ b) $\frac{1}{y^2}$ c) $\frac{1}{xy}$ d) $\frac{1}{(x+y)}$

1.2.75 Which ONE of the following is NOT a solution of the differential equation $\frac{d^2y}{dx^2}+y=1$? (CH 2008)

a) $y = 1$ b) $y = 1 + \cos x$ c) $y = 1 + \sin x$ d) $y = 2+\sin x+\cos x$

1.2.76 The limit of $\frac{\sin x}{x}$ as $x \to \infty$ is (CH 2008)

a) -1 b) 0 c) 1 d) ∞

1.2.77 A nonlinear function $f(x)$ is defined in the interval $-1.2 < x < 4$. The equation $f(x) = 0$ is solved for x using the Newton-Raphson iterative scheme. Among the initial guesses (I_1, I_2, I_3 and I_4), the guess that is likely to lead to the root most rapidly is

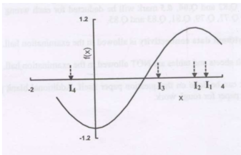


Fig. 1.2.77.1

(CH 2008)

a) I_1 b) I_2 c) I_3 d) I_4

1.2.78 Using Simpson's 1/3 rule and FOUR equally spaced intervals ($n = 4$), estimate the value of the integral $\int_0^{\frac{\pi}{4}} \frac{\sin x}{\cos^3 x}\, dx$ (CH 2008)

a) 0.3887 b) 0.4384 c) 0.5016 d) 0.5527

1.2.79 The following differential equation is to be solved numerically by the Euler's explicit method.

$$\frac{dy}{dx} = x^2y - 1.2y \text{ with } y(0) = 1$$

A step size of 0.1 is used. The solution for y at $x = 0.1$ is (CH 2008)

a) 0.880 b) 0.905 c) 1.000 d) 1.100

1.2.80 A reactor has been installed at a cost of Rs. 50,000 and is expected to have a working life of 10 years with a scrap value of Rs. 10,000. The capitalized cost (in Rs.) of the reactor based on an annual compound interest rate of 5% is (CH 2008)

a) $1,13,600$ b) $42,000$ c) $52,500$ d) $10,500$

1.2.81 $\lim_{x\to\infty} \frac{x-\sin x}{x+\cos x}$ equals (CS 2008)

a) 1 b) -1 c) ∞ d) 0

1.2.82 The minimum number of equal length subintervals needed to approximate $\int_1^2 xe^x\,dx$ to an accuracy of at least $\frac{1}{3}\times 10^{-6}$ using the trapezoidal rule is (CS 2008)

a) 1000e b) 1000 c) 100e d) 100

1.2.83 The Newton-Raphson iteration $x_{n+1} = \frac{1}{2}\left(x_n + \frac{R}{x_n}\right)$ can be used to compute the (CS 2008)

a) square of R
b) reciprocal of R
c) square root of R
d) logarithm of R

1.2.84 Let

$$P = \sum_{\substack{1\le i\le 2k \\ i \text{ odd}}} i \quad \text{and} \quad Q = \sum_{\substack{1\le i\le 2k \\ i \text{ even}}} i$$

where k is a positive integer. Then (CS 2008)

a) P=Q-K b) P=Q+K c) P=Q d) P=Q+2K

1.2.85 Let x_n denote the number of binary strings of length n that contain no consecutive 0s. (CS 2008)

a) Which of the following recurrences does x_n satisfy?

i) $x_n = 2x_{n-1}$ ii) $x_n = x_{\lfloor n/2\rfloor} + 1$ iii) $x_n = x_{\lfloor n/2\rfloor} + n$ iv) $x_n = x_{n-1} + x_{n-2}$

b) The value of x_5 is

i) 5 ii) 7 iii) 8 iv) 16

1.2.86 Which of the following is a solution to the differential equation $\frac{dx(t)}{dt} + 3x(t) = 0$? (EC 2008)

a) $x(t) = 3e^{-t}$ b) $x(t) = 2e^{-3t}$ c) $x(t) = \frac{3}{2}t^2$ d) $x(t) = 3t^2$

1.2.87 The recursion relation to solve $x = e^x$ using Newton-Raphson method is (EC 2008)

a) $x_{n+1} = e^{-x_n}$
b) $x_{n+1} = x_n - e^{-x_n}$
c) $x_{n+1} = (1+x_n)\frac{e^{-x_n}}{1+e^{-x_n}}$
d) $x_{n+1} = \frac{x_n^2 e^{-x_n}(1+x_n)-1}{x_n - e^{-x_n}}$

1.2.88 The value of the integral of the function $g(x,y) = 4x^3 + 10y$ along the straight line segment from the point $(0,0)$ to the point $(1,2)$ in the $x-y$ plane is (EC 2008)

a) 33 b) 35 c) 40 d) 56

1.2.89 A discrete time linear shift-invariant system has an impulse response $h[n]$ with $h[0] = 1$, $h[1] = -1$, $h[2] = 2$, and zero otherwise. The system is given an input sequence $x[n]$ with $x[0] = x[2] = 1$, and zero otherwise. The number of nonzero samples in the output sequence $y[n]$, and the value of $y[2]$ are, respectively (EC 2008)

a) $5, 2$ b) $6, 2$ c) $6, 1$ d) $5, 3$

1.2.90 Consider points P and Q in the x-y plane, with $P = (1, 0)$ and $Q = (0, 1)$. The line integral

$$2\int_P^Q (x\,dx + y\,dy)$$

along the semicircle with the line segment PQ as its diameter (EC 2008)

a) is -1
b) is 0
c) is 1
d) depends on the direction (clockwise or anti-clockwise) of the semicircle

1.2.91 $\{x(n)\}$ is a real-valued periodic sequence with a period N. $x(n)$ and $X(k)$ form N-point Discrete Fourier Transform (DFT) pairs. The DFT $Y(k)$ of the sequence

$$y(n) = x(r)x(n + r)$$

is (EC 2008)

a) $|X(k)|^2$
b) $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X^*(k + r)$
c) $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X(k + r)$
d) 0

1.2.92 In the following network, the switch is closed at $t = 0$ and the sampling starts from $t = 0$. The sampling frequency is 10 Hz. (EC 2008)

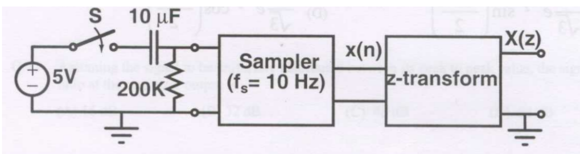


Fig. 1.2.92.1

a) The samples $x(n), n = 0, 1, 2, \ldots$ are given by

i) $5\left(1 - e^{-0.05n}\right)$ ii) $5e^{-0.05n}$ iii) $5\left(1 - e^{-5n}\right)$ iv) $5e^{-5n}$

b) The expression and the region of convergence of the z-transform of the sampled signal are

i) $\frac{5z}{z-e^5}$; $|z| < e^5$

ii) $\frac{5z}{z-e^{-0.05}}$; $|z| < e^{-0.05}$

iii) $\frac{5z}{z-e^{-0.05}}$; $|z| > e^{-0.05}$

iv) $\frac{5z}{z-e^5}$; $|z| > e^5$

1.2.93 A function $y(t)$ satisfies the following differential equation:

$$\frac{dy(t)}{dt} + y(t) = \delta(t)$$

where $\delta(t)$ is the delta function. Assuming zero initial condition, and denoting the unit step function by $u(t)$, $y(t)$ can be of the form (EE 2008)

a) e^t b) e^{-t} c) $e^t u(t)$ d) $e^{-t}u(t)$

1.2.94 Fourier analysis matches the signal by a series of sinusoids. Each member of the series fits an exact number of (GG 2008)

a) one-fourth wavelength
b) one-third wavelength
c) half-wavelength
d) one wavelength

1.2.95 Two sampled data sets are given as (GG 2008)

$$X(t) = \left[1, \frac{1}{2}, -1, -\frac{1}{2}\right]$$
$$Y(t) = \left[1, -\frac{1}{2}, -1, \frac{1}{2}\right].$$

a) The cross-correlation between these two time series for zero lag is

i) $-\frac{3}{2}$ ii) $\frac{5}{2}$ iii) $-\frac{1}{2}$ iv) $\frac{3}{2}$

b) The convolution of the data sets results in a time series (GG 2008)

i) $\left[1, 1, -\frac{11}{2}, 4, -\frac{17}{2}, -1, 1\right]$

ii) $\left[-1, 1, 4, \frac{17}{2}, -\frac{11}{2}, 1, 1\right]$

iii) $\left[1, \frac{17}{2}, 4, -\frac{11}{2}, -1, \frac{3}{2}, 1\right]$

iv) $\left[1, -1, \frac{17}{2}, 5, \frac{11}{2}, -1, 1\right]$

1.2.96 The fundamental period of the discrete-time signal $x[n] = e^{j\left(\frac{5\pi}{6}\right)n}$ is (IN 2008)

a) $\frac{6}{5\pi}$ b) $\frac{12}{5}$ c) 6 d) 12

1.2.97 Which one of the following discrete-time systems is time invariant? (IN 2008)

a) $y[n] = nx[n]$ b) $y[n] = x[3n]$ c) $y[n] = x[-n]$ d) $y[n] = x[n-3]$

1.2.98 Consider the differential equation $\frac{dy}{dx} = 1 + y^2$. Which one of the following can be a particular solution of this differential equation? (IN 2008)

a) $y = \tan(x+3)$ b) $y = \tan x + 3$ c) $x = \tan(y+3)$ d) $x = \tan y + 3$

1.2.99 It is known that two roots of the nonlinear equation $x^3 - 6x^2 + 11x - 6 = 0$ are 1 and 3. The third root will be (IN 2008)

a) j b) $-j$ c) 2 d) 4

1.2.100 Consider a discrete-time LTI system with input $x[n] = \delta[n] + \delta[n-1]$ and impulse response $h[n] = \delta[n] - \delta[n-1]$. The output of the system will be given by (IN 2008)

a) $\delta[n] - \delta[n-2]$
b) $\delta[n] - \delta[n-1]$
c) $\delta[n-1] + \delta[n-2]$
d) $\delta[n] + \delta[n-1] + \delta[n-2]$

1.2.101 Consider a discrete-time system for which the input $x[n]$ and the output $y[n]$ are related as $y[n] = x[n]$-$\frac{1}{3}y[n-1]$. If $y[n] = 0$ for $n < 0$ and $x[n] = \delta[n]$, then $y[n]$ can be expressed in terms of the unit step $u[n]$ (IN 2008)

a) $\left(\frac{-1}{3}\right)^n u[n]$ b) $\left(\frac{1}{3}\right)^n u[n]$ c) $(3)^n u[n]$ d) $(-3)^n u[n]$

1.2.102 The region of convergence of the z-transform of the discrete-time signal $x[n] = 2^n$ $u[n]$ will be (IN 2008)

a) $|z| > 2$ b) $|z| < 2$ c) $|z| > \frac{1}{2}$ d) $|z| < \frac{1}{2}$

1.2.103 $\lim_{x\to 0} \frac{\sin x}{x}$ is (IN 2008)

a) indeterminate b) 0 c) 1 d) ∞

2 Optimization

2.0.1 The minimum value of

$$J(x) = x^2 - 7x + 30$$

occurs at (AE 2007)

a) $x = 7/2$ b) $x = 7/30$ c) $x = 30/7$ d) $x = 30$

2.0.2 Consider the function $f(x) = x^2 - x - 2$. The maximum value of $f(x)$ in the closed interval $[-4, 4]$ is: (EC 2007)

a) 18 b) 10 c) -2.25 d) Indeterminate

2.0.3 For real x, the maximum value of $\dfrac{e^{\sin x}}{e^{\cos x}}$ is: (IN 2007)

a) 1
b) e
c) $e^{\sqrt{2}}$
d) ∞

2.0.4 Consider the function $f(x) = |x|^3$, where x is real. Then the function $f(x)$ at $x = 0$ is (IN 2007)

a) continuous but not differentiable
b) once differentiable but not twice
c) twice differentiable but not thrice
d) thrice differentiable

2.0.5 Consider the linear programming problem

$$\text{Maximize } z = c_1x_1 + c_2x_2, \quad c_1, c_2 > 0,$$
$$\text{subject to}$$
$$x_1 + x_2 \leq 3$$
$$2x_1 + 3x_2 \leq 4$$
$$x_i \geq 0$$

Then: (MA 2007)

a) the primal has an optimal solution but the dual does NOT have an optimal solution
b) both the primal and the dual have optimal solutions
c) the dual has an optimal solution but the primal does NOT have an optimal solution
d) neither the primal nor the dual have optimal solutions

2.0.6 For each $a \in \mathbb{R}$, consider the linear programming problem:

$$\text{Max. } z = x_1 + 2x_2 + 3x_3 + 4x_4$$
subject to (MA 2007)

$$ax_1 + 2x_2 \leq 1$$
$$x_1 + 2x_2 + 3x_3 \leq 2$$
$$x_1, x_2, x_3, x_4 \geq 0$$

Let $S = \{a \in \mathbb{R} : \text{the given LP problem has a basic feasible solution}\}$. Then:

a) $S = \emptyset$
b) $S = \mathbb{R}$
c) $S = (0, \infty)$
d) $S = (-\infty, 0)$

2.0.7 Consider the linear programming problem:
Max. $z = x_1 + 5x_2 + 3x_3$ subject to

$$2x_1 - 3x_2 + 5x_3 \leq 3$$
$$x_1 - x_2 \leq 5$$
$$x_1, x_2, x_3 \geq 0$$

Then the dual of this LP problem: (MA 2007)

a) has a feasible solution but does NOT have a basic feasible solution

b) has a basic feasible solution
c) has infinite number of feasible solutions
d) has no feasible solution

2.0.8 Consider a transportation problem with two warehouses and two markets. The warehouse capacities are $a_1 = 2$ and $a_2 = 4$, and the market demands are $b_1 = 3$ and $b_2 = 3$. Let x_{ij} be the quantity shipped from warehouse i to market j, and c_{ij} be the corresponding unit cost. Suppose that $c_{11} = 1$, $c_{21} = 1$, and $c_{22} = 2$. Then $(x_{11}, x_{12}, x_{21}, x_{22}) = (2, 0, 1, 3)$ is optimal for every: (MA 2007)

a) $c_{12} \in [1, 2]$
c) $c_{12} \in [1, 3]$
b) $c_{12} \in [0, 3]$
d) $c_{12} \in [2, 4]$

2.0.9 The minimum value of function $y = x^2$ in the interval $[1, 5]$ is (ME 2007)

a) 0 b) 1 c) 25 d) undefined

2.0.10 Three jobs A, B, and C are to be assigned to three machines X, Y and Z. The processing costs are given below. The minimum total cost of assigning the jobs to the machines is (MN 2007)

		Machine		
Job	A	19	28	31
	B	11	17	16
	C	12	15	13

TABLE 2.0.10

a) 60 b) 54 c) 51 d) 49

2.0.11 Consider the following linear programming problem: Maximize $Z = 6X_1 + 4X_2$ Subject to

$$2X_1 \leq 8,$$
$$2X_2 \leq 12,$$
$$3X_1 + 2X_2 \leq 18,$$
$$X_1 \geq 0, \; X_2 \geq 0$$

The multiple optimal solutions lie on the line joining the corner points (MN 2007)

a) (0, 0), (0, 6) b) (0, 6), (2, 6) c) (2, 6), (4, 3) d) (4, 3), (4, 0)

2.0.12 A product is made by mixing three raw materials 1, 2, 3 in varying proportions, where material I must account for not more than 50% of the total. If x, y and z are the amounts of raw materials 1, 2, 3 respectively, this constant can be modeled as (2007 PI)

a) $x \leq 0.5$ b) $x \leq 0.5(x+y+z)$ c) $0.5x \leq x+y+z$ d) $x \geq 0.5(y+z)$

2.0.13 For the function $f(x,y) = x^2 - y^2$ defined in R^2, the point [0,0] is (2007 PI)

a) a local minimum
b) a local maximum
c) neither a local minimum nor a local maximum
d) both a local maximum and a local minimum

2.0.14 In an optimization problem, let y be a 0-1 variable and x be a positive real number. Now, the condition that x can take non-zero values only if $y = 1$ can be modeled using the linear constraint

(2007 PI)

a) $x \leq My$ (M is a large number) c) $x \geq My$ (M is a large number)
b) $x \geq y$ d) $xy \geq 0$

2.0.15 Item P is made from components Q and R. Item Q, in turn, is made from S and T. The lead times for items P, Q, R, S, and T are 2, 3, 10, 5, and 6 weeks, respectively. The lead time (in weeks) needed to respond to a customer order for item P is

(2007 PI)

a) 10 b) 11 c) 12 d) 26

2.0.16 Four jobs have to be sequenced on a single facility, with the objective of minimizing the maximum tardiness

$$\max_i \left|\text{Completion time}_i - \text{Due date}_i\right|.$$

The jobs have due dates and processing times as follows:

Job	Due date (day number)	Processing time (days)
P	5	2
Q	6	10
R	3	3
S	7	4

The last job that should be taken up is (2007 PI)

a) P b) Q c) R d) S

2.0.17 An asset investment is made for Rs. 1,20,000. The uniform costs per year are Rs. 40,000 in operating the asset. Uniform benefits per year are either Rs. 60,000 or Rs. 80,000, judged to be equally likely. What is the expected payback period? (2007 PI)

a) 3 b) 4.5 c) 6 d) 9

2.0.18 For the data given below, how many machines can be assigned to an operator to minimize idle time of the operator and machines? (2007 PI)

Activity	Time (minutes)
machine loading + unloading	2
machining	4
walking from one machine to the next	1

a) 1 b) 2 c) 3 d) 4

2.0.19 The problem of finding the rectangle of maximum area with perimeter equal to 20 can be posed as the constrained optimization problem

$$\begin{aligned} \text{Max} \quad & xy \\ \text{s.t.} \quad & 2x + 2y = 20 \\ & x, y \geq 0 \end{aligned}$$

The solution to this problem is $x = y = 5$. What is the value of the Lagrange multiplier corresponding to the perimeter constraint? (2007 PI)

a) 2.5 b) 5 c) 7.5 d) 10

2.0.20 A manufacturing system with a production rate p units/day experiences a demand rate of d units/day where p ¿ d. Let Q be the maximum production quantity per period. When the total production in a period reaches Q units, the production is stopped and restarted only when inventory becomes zero. In such a scenario, the maximum cycle inventory is (2007 PI)

a) $Q \cdot p \cdot (p - d)$ b) $\frac{Q}{(p-d)^p}$ c) $\frac{Q}{p}(p - d)$ d) $\frac{p(p-d)}{Q}$

2.0.21 In a time study, the observed times and ratings for an elemental operation are as shown below:

	Reading 1	Reading 2
Rating (%)	80	100
Observed time (minutes)	0.60	0.50

Considering an allowance of 10% of the normal time, the standard time (in minutes) for the operation is:

(2007 PI)

a) 0.49 b) 0.54 c) 0.98 d) 1.08

2.0.22 The figure below illustrates a project network describing the precedence relationships among different activities (A-J). The activities along with their duration in weeks are represented as arcs, and the events are shown as nodes (1 is the start event and 9 is the end event). (2007 PI)

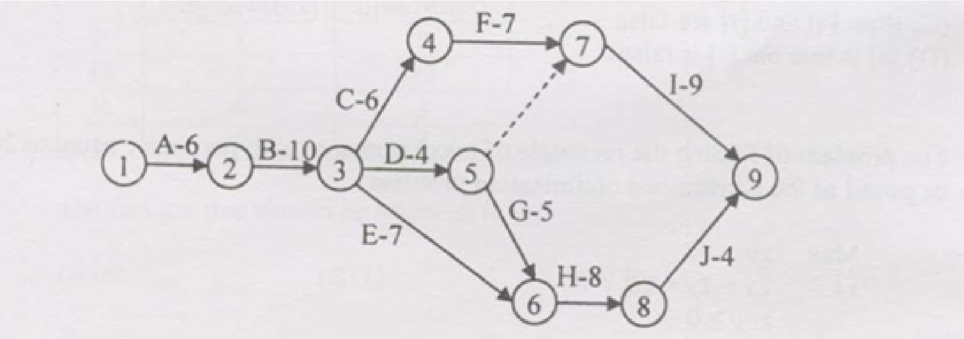


Fig. 2.0.22.1

a) The length of the crucial path in weeks is

i) 29 ii) 31 iii) 38 iv) 66

b) If U_α is the earliest start time of event α, then the reccurence equation defining U_6

i) $U_6 = \text{Max}\{U_8, 8\}$
ii) $U_6 = U_8 - 8$
iii) $U_6 = \text{Max}\{U_3, U_5, 7, 5\}$
iv) $U_6 = \text{Max}\{U_3 + 7, U_5 + 5\}$

c) If activity B has uncertain duration and is uniformly distributed over the interval [8, 12], and T is the earliest start time of event 3 (assume that event I starts at time 0), then the mean and variance of T are

i) 10 and 0.4 ii) 10 and 1.33 iii) 16 and 0.4 iv) 16 and 1.33

2.0.23 The maximum value of the function 2x + 3y + 4z on the ellipsoid $2x^2+3y^2+4z^2 = 1$ is (2007 XE)

a) 2 b) 3 c) 6 d) 9

2.0.24 The function $f(x) = x^2 - 5x + 6$ (AE 2008)

a) has its maximum value at $x = 2.0$
b) has its maximum value at $x = 2.5$
c) is increasing on the interval (2.0, 2.5)
d) is increasing on the interval (2.5, 3.0)

2.0.25 If $\log_e(y) = -x\log_e(x)$, then the maximum value of y is (AG 2008)

a) e b) e^{x^2} c) $e^{e^{-1}}$ d) e^x

2.0.26 If $\sum_{i=1}^{n}(x - a_i)^2$ has a minima at A, then A is the arithmetic mean of the series (AG 2008)

a) $a_1 - a_2 + a_3 - \ldots + (-1)^{n+1} a^n$
b) $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots + \frac{1}{a_n}$
c) $\frac{1}{a_1} - \frac{1}{a_2} + \frac{1}{a_3} - \ldots + (-1)^{n+1} \frac{1}{a_n}$
d) $a_1 + a_2 + a_3 + \ldots + a_n$

2.0.27 A point on a curve is said to be an extremum if it is a local minimum or a local maximum. The number of distinct extrema for the curve $3x^4 - 16x^3 + 24x^2 + 37$ is (CS 2008)

a) 0 b) 1 c) 2 d) 3

2.0.28 Aishwarya studies either computer science or mathematics everyday. If she studies computer science on a day, then the probability that the studies mathematics the next day is 0.6. If she studies mathematics on a day, then the probability that the studies computer science the next day is 0.4. Given that Aishwarya studies computer science on Monday, what is the probability that she studies computer science on Wednesday? (CS 2008)

a) 0.24 b) 0.36 c) 0.4 d) 0.6

2.0.29 For real values of x, the minimum value of the function $f(x) = \exp(x) + \exp(-x)$ is (EC 2008)

a) 2 b) 1 c) 0.5 d) 0

2.0.30 Given y = x^2 + 2x + 10 , the value of $\left.\frac{dy}{dx}\right|_{x=1}$ is equal to (IN 2008)

a) 0 b) 4 c) 12 d) 13

2.0.31 The expression $e^{-\ln x}$ for $x > 0$ is equal to (IN 2008)

a) $-x$ b) x c) x^{-1} d) $-x^{-1}$

2.0.32 Consider the function $y = x^2 - 6x + 9$.The maximum value of y obtained when x varies over the interval 2 to 5 is (IN 2008)

a) 1 b) 3 c) 4 d) 9

3 Probability

3.0.1 For station X, the maximum one day rainfall with 25 years return period is 100 mm. The probability of a one day rainfall equal to or greater than 100 mm at station X occurring at least once in 15 successive years is (AG 2007)

a) 0.458 b) 0.500 c) 0.637 d) 0.990

3.0.2 If the standard deviation of the spot speed of vehicles in a highway is $8.8 kmph$ and the mean speed of the vehicles is $33 kmph$, the coefficient of variation in speed is (CE 2007)

a) 0.1517 b) 0.1867 c) 0.2666 d) 0.3646

3.0.3 Suppose we uniformly and randomly select a permutation from the 20! permutations of $1, 2, 3, ..., 20$. What is the probability that 2 appears at an earlier position than any other even number in selected permutation ? (CS 2007)

a) $\frac{1}{2}$ b) $\frac{1}{10}$ c) $\frac{9!}{20!}$ d) None of these

3.0.4 There are n stations in a slotted LAN. Each station attempts to transmit with a probability p in each time slot. What is the probability that ONLY one station transmits in a given time slot? (CS 2007)

a) $np(1-p)^{n-1}$ b) $(1-p)^{n-1}$ c) $p(1-p)^{n-1}$ d) $1-(1-p)^{n-1}$

3.0.5 If E denotes expectation, the variance of a random variable X is given by: (EC 2007)

a) $E[X^2] - E^2[X]$
b) $E[X^2] + E^2[X]$
c) $E[X^2]$
d) $E^2[X]$

3.0.6 If $R(\tau)$ is autocorrelation of real, WSS random process, which is NOT true: (EC 2007)

a) $R(\tau) = R(-\tau)$
b) $|R(t)| \leq R(0)$
c) $R(\tau) = -R(-\tau)$
d) Mean square value is $R(0)$

3.0.7 If $S(f)$ is power spectral density of a real WSS random process, which is ALWAYS true:

(EC 2007)

a) $S(0) \geq S(f)$ b) $S(f) \geq 0$ c) $S(-f) = -S(f)$ d) $\int S(f)df = 0$

3.0.8 An examination consists of two papers, Paper I and Paper II. The probability of failing in Paper I is 0.3 and that in Paper II is 0.2. Given that a student has failed in Paper II, the probability of failing in Paper I is 0.6. The probability of a student failing in both the papers is: (EC 2007)

a) 0.5 b) 0.18 c) 0.12 d) 0.06

3.0.9 During transmission over a binary channel, bit errors occur independently with probability p. The probability of at most one bit in error in a block of n bits is: (EC 2007)

a) p^n
b) $1 - p^n$
c) $np(1-p)^{n-1} + (1-p)^n$
d) $1-(1-p)^n$

3.0.10 Two 4-ary signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis and four symbols in each are equiprobable. Let $N_0/2$ denote the power spectral density of white Gaussian noise. (EC 2007)

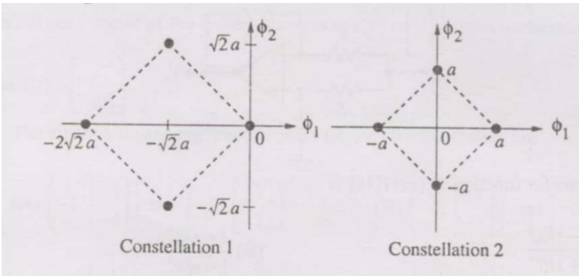


Fig. 3.0.10.1

a) The ratio of the average energy of Constellation 1 to that of Constellation 2 is:

i) $4a^2$ ii) 4 iii) 2 iv) 8

b) If these constellations are used over an AWGN channel, then:

i) Probability of symbol error for Constellation 1 is lower
ii) Probability of symbol error for Constellation 1 is higher
iii) Probability of symbol error is equal for both
iv) Value of N_0 will determine which has lower error

3.0.11 An input to a 6-level quantizer has PDF $f(x)$ as shown: it is piecewise-constant with three decision boundaries at -1, 0, and 1, chosen to maximize output entropy. The central plateau height is a, the outer flat region height is b. (EC 2007)

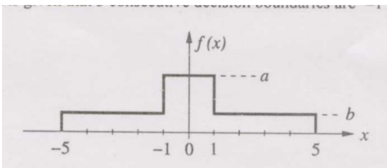


Fig. 3.0.11.1

a) The values of a and b are:

i) $a = \frac{1}{6},\ b = \frac{1}{12}$ ii) $a = \frac{1}{5},\ b = \frac{3}{40}$ iii) $a = \frac{1}{4},\ b = \frac{1}{16}$ iv) $a = \frac{1}{3},\ b = \frac{1}{24}$

b) Assuming reconstruction levels are midpoints of decision intervals, the ratio of signal power to quantization noise power is:

i) $\frac{152}{9}$ ii) $\frac{64}{3}$ iii) $\frac{76}{3}$ iv) 28

3.0.12 A loaded dice has following probability distribution of occurrences

Dice value	1	2	3	4	5	6
Probability	1/4	1/8	1/8	1/8	1/8	1/4

If three identical dice as the above are thrown, the probability of occurrence of values 1, 5 and 6 on the three dice is (EE 2007)

a) same as that of occurrence of 3, 4, 5
b) same as that of occurrence of 1, 2, 5
c) 1/128
d) 5/8

3.0.13 A coin is tossed thrice. Let X be the event that head occurs in each of the first two tosses. Let Y be the event that a tail occurs on the third toss. Let Z be the event that two tails occur in three tosses. Based on the above information, which one of the following statements is TRUE? (GE 2007)

a) X and Y are not independent
b) Y and Z are dependent
c) Y and Z are independent
d) X and Z are independent

3.0.14 Assume that the duration in minutes of a telephone conversation follows the exponential distribution

$$f(x) = \frac{1}{5}e^{-x/5}, x \geq 0$$

. The probability that the conversation will exceed five minutes is (IN 2007)

a) $\frac{1}{e}$ b) $\frac{1}{5}$ c) $\frac{1}{3}$ d) $1 - \frac{1}{e}$

3.0.15 Two sensors have measurement errors that are Gaussian distributed with zero means and variances σ_1^2 and σ_2^2, respectively. The two sensor measurements x_1 and x_2 are combined to form the weighted average $x = \alpha x_1 + (1 - \alpha)x_2$, $0 \leq \alpha \leq 1$. Assuming that the measurement errors of the two sensors are uncorrelated, the weighting factor α that yields the smallest error variance of x is (IN 2007)

a) $\frac{\sigma_1^2}{\sigma_1^2+\sigma_2^2}$ b) $\frac{\sigma_2^2}{\sigma_1^2+\sigma_2^2}$ c) $\frac{\sigma_1}{\sigma_1+\sigma_2}$ d) 0.5

3.0.16 The value of the integral $\int_0^\infty \int_0^\infty e^{-x^2-y^2}\, dx\, dy$ is (IN 2007)

a) $\sqrt{\pi}/2$ b) $\sqrt{\pi}$ c) π d) $\pi/4$

3.0.17 Let X, Y be jointly distributed random variables having the joint probability density function

$$f(x, y) = \begin{cases} 1, & \text{if } 0 < x + y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Then $P(Y \geq \max(X, 1 - X))$ is (MA 2007)

a) $\frac{1}{2}$ b) 1 c) $\frac{1}{4}$ d) $\frac{1}{6}$

3.0.18 Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed chi-square random variables, each having 4 degrees of freedom. Define

$$S_n = \sum_{i=1}^{n} X_i$$

If $\frac{S_n}{n} \to \mu$ as $n \to \infty$, then $\mu =$ (MA 2007)

a) 8 b) 16 c) 24 d) 32

3.0.19 Let A, B and C be three events such that:

$$P(A) = 0.4, P(B) = 0.5, P(A \cup B) = 0.6, P(C) = 0.6 \text{and } P(A \cap B \cap C^c) = 0.1.$$

Then $P(A \cap B \cap C) =$ (MA 2007)

a) $\dfrac{1}{2}$ b) $\dfrac{1}{3}$ c) $\dfrac{1}{4}$ d) $\dfrac{1}{5}$

3.0.20 Consider two identical boxes B_1 and B_2, where the box B_i $(i = 1, 2)$ contains $i+1$ red and $5-i+1$ white balls. A fair die is cast. Let the number of dots shown on the top face of the die be N. If N is even or 5, then two balls are drawn with replacement from the box B_1; otherwise, two balls are drawn with replacement from the box B_2. The probability that the two drawn balls are of different colours is: (MA 2007)

a) $\dfrac{7}{25}$ b) $\dfrac{9}{25}$ c) $\dfrac{12}{25}$ d) $\dfrac{16}{25}$

3.0.21 Let $X_1, X_2, \ldots$ be a sequence of independent and identically distributed random variables with

$$P(X_i = 1) = P(X_i = -1) = \frac{1}{2}.$$

Suppose for the standard normal random variable Z, $P(-0.1 < Z \leq 0.1) = 0.08$. If $S_n = \sum_{i=1}^{n} X_i$, then (MA 2007)

$$\lim P\left(\frac{S_n}{\sqrt{n}} > \frac{n}{10}\right) =$$

a) 0.42 b) 0.46 c) 0.5 d) 0.54

3.0.22 Let $X_1, X_2, \ldots, X_5$ be a random sample of size 5 from a population having standard normal distribution. Let

$$\bar{X} = \frac{1}{5}\sum_{i=1}^{5} X_i \quad \text{and} \quad T = \sum_{i=1}^{5}(X_i - \bar{X})^2.$$

Then $E(T^2\bar{X}^2) =$ (MA 2007)

a) 3 b) 3.6 c) 4.8 d) 5.2

3.0.23 Let $x_1 = 3.5$, $x_2 = 7.5$ and $x_3 = 5.2$ be observed values of a random sample of size three from a population having uniform distribution over the interval $(\theta, \theta+5)$, where $\theta \in (0, \infty)$ is unknown and is to be estimated. Then which of the following is NOT a maximum likelihood estimate of θ? (MA 2007)

a) 2.4 b) 2.7 c) 3 d) 3.3

3.0.24 Let X and Y be jointly distributed random variables such that the conditional distribution of Y, given $X = x$, is uniform on the interval $(x - 1, x + 1)$. Suppose $\mathbb{E}(X) = 1$ and $\mathrm{Var}(X) = \frac{5}{3}$. (MA 2007)

a) The mean of the random variable Y is

i) $\frac{1}{2}$ ii) 1 iii) $\frac{3}{2}$ iv) 2

b) The variance of the random variable Y is

i) $\frac{1}{2}$ ii) $\frac{2}{3}$ iii) 1 iv) 2

3.0.25 Let X and Y be two independent random variables. Which one of the relations between expectation (E), variance (Var) and covariance (Cov) given below is FALSE? (ME 2007)

a) $E(XY) = E(X)E(Y)$
b) $\mathrm{Cov}(X, Y) = 0$
c) $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$
d) $E(X^2Y^2) = (E(X))^2(E(Y))^2$

3.0.26 The random variable X has the following probability mass function:

$$P(4) = \frac{1}{4}, \quad P(8) = \frac{1}{4}, \quad P(12) = \frac{1}{4}, \quad P(16) = \frac{1}{4}$$

The expected value of X is: (MN 2007)

a) 1 b) 3 c) 10 d) 12

3.0.27 The time between successive failures (in hours) of a side discharge loader operating in a mechanised underground coal mine are as follows:

62, 58, 54, 50, 52, 60, 58, 57, 50, 53

If the failure data follow an exponential distribution, then reliability of the equipment for a period of 50 hours is: (MN 2007)

a) 0.25 b) 0.40 c) 0.60 d) 1.00

3.0.28 The carbon concentration profile in decarburization is given by:

$$C(x,t) = L + M \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right),$$

where $x = 0.5\,mm$, $C_{initial} = 1.2\%$, $C_{final} = 0.8\%$, $D = 1.28 \times 10^{-11} m^2/s$. The approximate time t is (2007 MT)

a) 30 hours
b) 3 hours
c) 3 minutes
d) 30 seconds

3.0.29 The probability distribution function is $p(x) = \frac{1}{\sqrt{\pi}} e^{-x^2}$. Using the trapezoidal rule, the probability that x lies between 0.6 and 0.8 is (2007 MT)

a) 0 b) 0.069 c) 0.138 d) 0.56

3.0.30 Two cards are drawn at random in succession,with replacement, from a deck of 52 well shuffled cards. Probability of getting both 'Aces' is (2007 PI)

a) 1/169 b) 2/169 c) 1/13 d) 2/13

3.0.31 If X is a continuous random variable whose probability density function is given by

$$f(x) = \begin{cases} K(5x - 2x^2), & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Then $P(X > 1)$ is (2007 PI)

a) 3/14 b) 4/5 c) 14/17 d) 17/28

3.0.32 The random variable X takes on the values 1,2 or 3 with probabilities $(2+5P)/5, (1+3P)/5$, and $(1.5+2P)/5$,respectively. The values of P and $E[X]$ are respectively (2007 PI)

a) 0.05,1.87 b) 1.90,5.87 c) 0.05,1.10 d) 0.25,1.40

3.0.33 The average number of accidents occurring monthly on an assembly shop floor is 2. The probability that there will be at least one accident in this month is estimated to be (2007 PI)

a) 0.055 b) 0.456 c) 0.865 d) 0.950

3.0.34 $X_1, \ldots, X_{100}$ are Bernoulli random variables with a probability of success equal to 0.6. By the Central Limit Theorem, the random variable

$$Y = \sum_{i=1}^{100} X_i$$

is approximately normally distributed. Then Y has mean and variance respectively equal to (2007 PI)

a) 40 and 24 b) 60 and 24 c) 40 and 12 d) 60 and 12

3.0.35 A process is to be controlled with standard values $\mu = 15$ and $\sigma = 3.6$. The sample size is 9. The control limits for the X chart are (2007 PI)

a) 15 ± 10.8 b) 15 ± 3.6 c) 0.4 ± 10.8 d) 0.4 ± 3.6

3.0.36 The reliability of an equipment for a time to failure exceeding t is given by

$$R(t) = \exp(-\lambda t)$$

The mean time to failure (MTTF) for this equipment (in hours) is (2007 PI)

a) λ b) $\frac{1}{\lambda}$ c) $\frac{1}{\lambda^2}$ d) λ^2

3.0.37 Two students take a test consisting of five TRUE/FALSE questions. To pass the test the students have to answer at least three questions correctly. Both of them know the correct answers to two questions and guess the answers to the remaining three. The probability that only one student passes the test is equal to: (2007 XE)

a) $\frac{6}{32}$ b) $\frac{7}{32}$ c) $\frac{1}{4}$ d) $\frac{3}{4}$

3.0.38 Choose a point uniformly at random on the disc $x^2 + y^2 \leq 1$. Let the random variable X denote the distance of this point from the center of the disc. Then the variance of X is: (2007 XE)

a) $\frac{1}{16}$ b) $\frac{1}{17}$ c) $\frac{1}{18}$ d) $\frac{1}{19}$

3.0.39 If $f(x)$ is a perfect normal distribution with mean and standard deviation of 5 and 1 respectively, then the value of $f(x)$ for $x = 6$ is (AG 2008)

a) 0.124 b) 0.242 c) 0.482 d) 0.524

3.0.40 If the probability density function of a random variable X is

$$f(x) = \begin{cases} x^2, & \text{for } -1 \leq x \leq 1 \\ 0, & \text{for any other value of } x \end{cases}$$

then, the percentage probability $P\left(-\frac{1}{3} \leq x \leq \frac{1}{3}\right)$ is: (CE 2008)

a) 0.247
b) 2.47
c) 24.7
d) 247

3.0.41 A person on a trip has a choice between private car and public transport. The probability of using a private car is 0.45. While using the public transport, further choices available are bus and metro, out of which the probability of commuting by a bus is 0.55. In such a situation, the probability (*rounded up to two decimals*) of using a car, bus and metro, respectively would be: (CE 2008)

a) 0.45, 0.30 and 0.25
b) 0.45, 0.25 and 0.30
c) 0.45, 0.55 and 0.00
d) 0.45, 0.35 and 0.20

3.0.42 The normal distribution is given by

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

The points of inflexion to the normal curve are (CH 2008)

a) $x = -\sigma, +\sigma$
b) $x = \mu + \sigma, \mu - \sigma$
c) $x = \mu + 2\sigma, \mu - 2\sigma$
d) $x = \mu + 3\sigma, \mu - 3\sigma$

3.0.43 The Poisson distribution is given by $P(r) = \frac{m^r}{r!} \exp(-m)$. The first moment about the origin for this distribution is (CH 2008)

a) 0
b) m
c) $1/m$
d) m^2

3.0.44 Let X be a random variable following normal distribution with mean +1 and variance 4. Let Y be another normal variable with mean −1 and variance unknown. If

$$P(X \le -1) = P(Y \ge 2),$$

the standard deviation of Y is (CS 2008)

a) 3
b) 2
c) $\sqrt{2}$
d) 1

3.0.45 The probability density function (PDF) of a random variable X is as shown below.

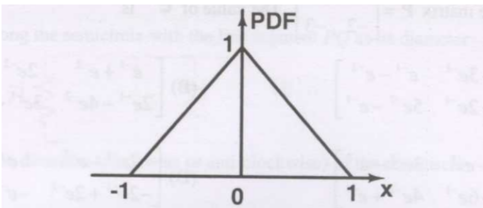


Fig. 3.0.45.1

The corresponding cumulative distribution function (CDF) has the form (EC 2008)

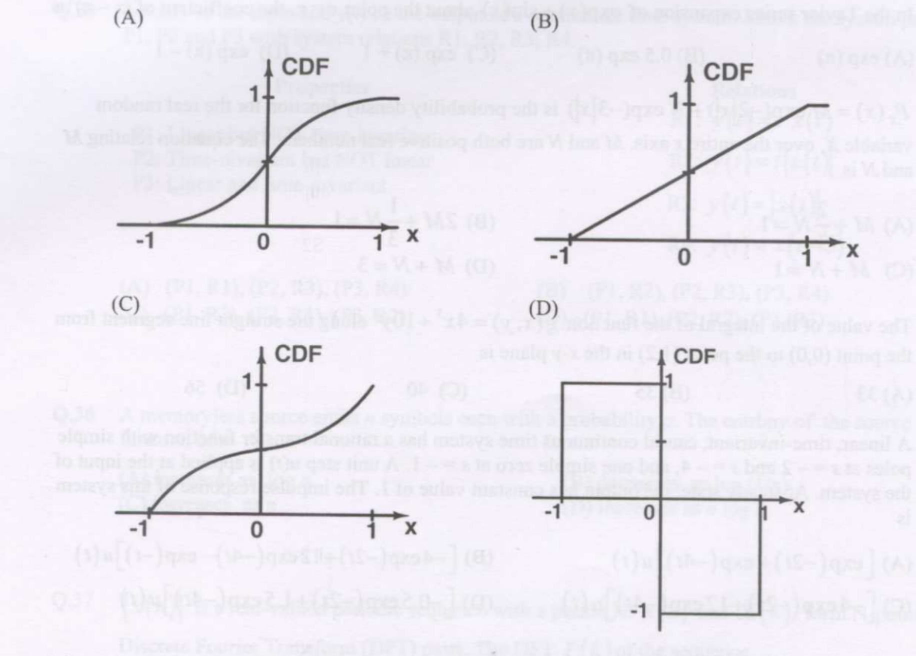


Fig. 3.0.45.2

3.0.46 $P_X(x) = M\exp(-2|x|) + N\exp(-3|x|)$ is the probability density function for the real random variable X, over the entire x axis. M and N are both positive real numbers. The equation relating M and N is (EC 2008)

a) $M + \frac{2}{3}N = 1$ b) $2M + \frac{1}{3}N = 1$ c) $M + N = 1$ d) $M + N = 3$

3.0.47 A memoryless source emits n symbols each with a probability p. The entropy of the source as a function of n (EC 2008)

a) increases as $\log n$
b) decreases as $\log(1/n)$
c) increases as n
d) increases as $n \log n$

3.0.48 Consider a Binary Symmetric Channel (BSC) with probability of error being p. To transmit a bit, say 1, we transmit a sequence of three 1s. The receiver will interpret the received sequence to represent 1 if at least two bits are 1. The probability that the transmitted bit will be received in error is (EC 2008)

a) $p^3 + 3p^2(1-p)$
b) p^3
c) $(1-p)^3$
d) $p^3 + p^2(1-p)$

3.0.49 A speech signal, band limited to 4 kHz and peak voltage varying between +5V and -5V, is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits. (EC 2008)

a) If the bits 0 and 1 are transmitted using bipolar pulses, the minimum bandwidth required for distortion free transmission is

i) 64 kHz ii) 32 kHz iii) 8 kHz iv) 4 kHz

b) Assuming the signal to be uniformly distributed between its peak to peak value, the signal to noise ratio at the quantizer output is

i) 16 dB ii) 32 dB iii) 48 dB iv) 64 dB

c) The number of quantization levels required to reduce the quantization noise by a factor of 4 would be

i) 1024 ii) 512 iii) 256 iv) 64

3.0.50 X is a uniformly distributed random variable that takes values between 0 and 1. The value of $E\left(X^3\right)$ will be (EE 2008)

a) 0 b) $\frac{1}{8}$ c) $\frac{1}{4}$ d) $\frac{1}{2}$

3.0.51 Consider a Gaussian distributed random variable with zero mean and standard deviation σ . The value of its cumulative distribution function at the origin will be (IN 2008)

a) 0 b) 0.5 c) 1 d) 10 σ

3.0.52 A random variable is uniformly distributed over the interval 2 to 10. Its variance will be (IN 2008)

a) $\frac{16}{3}$ b) 6 c) $\frac{256}{9}$ d) 36